Supplementary material: Birkhoff discretization and complex Hamiltonian residual control of a magnetic catheter model

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Abstract

This supplement provides the mechanics, endpoint and matrix derivations, controlled-map identities, learning-error bounds and reproduction protocols for the three-subproblem article. The same revised beam and endpoint magnet are used throughout. Detailed formulas distinguish the matrix implementation, residual compensation and finite-time controlled convergence from the additional hypotheses of discrete KAM. A separate final section retains learned-reference tracking as a related extension; it does not replace the compensation objective analyzed in the main article. Source provenance and a framework-to-section map accompany the reproducible project.

Keywords

supplementary derivations, parameter provenance, numerical verification, reproducibility

S1 Mechanical derivation and parameter traceability

S1.1 General rod reduction and force accounting

For a spatial rod, define reference strains and body angular velocity by

$$\Gamma = D^{\mathsf{T}} r_s, \qquad D_s = D\widehat{\kappa}, \qquad D_t = D\widehat{\omega}_b. \tag{S1}$$

The hat map satisfies $\widehat{a}b = a \times b$. The spatial balance laws are

$$\rho A_c r_{tt} = n_s + f_{\mathrm{mag}} + f_{\mathrm{h}} + f_{\mathrm{c}} + \rho A_c g, \tag{S2}$$

$$\partial_t (D\mathcal{J}_b \omega_b) = m_s + r_s \times n + \tau_{\mathrm{ext}}. \tag{S3}$$

Here $\mathcal{J}_b$ is the rotational inertia per unit reference length. For constant body inertia, the second left-hand side equals $D(\mathcal{J}_b \dot{\omega}_b + \omega_b \times \mathcal{J}_b \omega_b)$. A stored-energy density determines the unconstrained force and moment components through its strain derivatives (Simo 1985; Antman 2005). In an unshearable specialization, the eliminated shear strains have constraint reactions; setting those strains to zero in an unconstrained shear law does not set the transverse internal forces to zero.

A body-fixed moment density $m_b(s)$ in a static applied field has potential

$$V_{\mathrm{mag}} = -\int_0^L (Dm_b) \cdot B_0(r)\, \mathrm{d}s. \tag{S4}$$

The point-dipole benchmark replaces this distribution by a distal localized core. The permanent-dipole formulas are the leading terms of a localized magnetic stress resultant (Abbott et al. 2020). In a surrounding vacuum region, the Maxwell stress is

$$T_M = \mu_0^{-1} \left(BB^{\mathsf{T}} - \tfrac{1}{2}|B|^2 I \right). \tag{S5}$$

Its surface force and moment use the total field. Equivalently, for a complete localized steady current $j(\xi)$, including bound currents, and externally generated B_{ext},

$$\begin{aligned} m_d &= \tfrac{1}{2} \int \xi \times j(\xi)\, \mathrm{d}V, \\ F &= \nabla_r [m_d \cdot B_{\mathrm{ext}}(r)] + R_F, \\ \tau &= m_d \times B_{\mathrm{ext}}(r) + R_\tau. \end{aligned} \tag{S6}$$

Current closure gives $\int j\, \mathrm{d}V = 0$ and $\int \xi_i j_j\, \mathrm{d}V = \epsilon_{ijk}(m_d)_k$. Taylor expansion then yields, with $J_2 = \int |\xi|^2 |j(\xi)|\, \mathrm{d}V$,

$$|R_F| \le \tfrac{1}{2} J_2 \sup \|D^2 B_{\mathrm{ext}}\|, \qquad |R_\tau| \le J_2 \sup \|DB_{\mathrm{ext}}\|. \tag{S7}$$

The suprema cover the core and the Taylor segments. The self-field of the complete isolated element has no net self-force or self-torque. Thus one uses either the stress resultant or its dipole approximation, without counting both as separate loads. The approximation $\mu \simeq B_r a_m^3/\mu_0$ also assumes an effectively uniform, body-fixed magnetization; a grade specification is not a measured moment of an assembled tip.

Fluid and contact terms require their own constitutive choices. For example, a local added-mass and drag closure may be written as

$$f_{\mathrm{h}} = m_f a_f + m_a(a_f - r_{tt}) - c_1 v_r - c_2 |v_r| v_r, \qquad v_r = r_t - v_f, \qquad c_1, c_2 \geq 0. \tag{S8}$$

This is a reduced closure to be calibrated, not a resolved fluid interaction. In still fluid the drag power is nonpositive. In moving fluid it is dissipative relative to v_f and may supply energy to the rod. A consistently defined added mass may be incorporated into the kinetic metric once; it must then be removed from the external inertial load. For a stationary wall, penetration $\delta > 0$ and $n_c = \nabla\delta$ give one nonadhesive penalty example:

$$V_c = \frac{k_c}{p_c + 1}[\delta]_+^{p_c+1}, \qquad f_c = -k_c[\delta]_+^{p_c} n_c - c_c \mathbf{1}_{\{\delta>0\}}[\dot{\delta}]_+ n_c. \tag{S9}$$

The second term dissipates energy during compression. Such switching contact laws are excluded from the analytic neighbourhood used for the persistence results. Reduced gravity alone does not remove fluid forcing or select a blood-flow law. Neither (S8) nor (S9) is activated in the revised numerical benchmark.

For a configuration map satisfying the constraints, define

$$r_i = \partial_{q_i} r, \qquad \psi_i = \mathrm{vee}(D^{\mathsf{T}} \partial_{q_i} D). \tag{S10}$$

Then $\omega_b = \sum_i \psi_i \dot{q}_i$, and

$$M_{ij}(q) = \int_0^L [\rho A_c \, r_i \cdot r_j + \psi_i^{\mathsf{T}} \mathcal{J}_b \psi_j] \, \mathrm{d}s, \qquad Q_i = \int_0^L [f \cdot r_i + \tau \cdot D\psi_i] \, \mathrm{d}s + Q_i^{\partial}. \tag{S11}$$

Concentrated tip masses and their moments of inertia add their corresponding kinetic terms. A centreline interpolation by itself does not specify rotational inertia.

Variation of $T - V$ gives

$$(c_{\mathrm{geo}})_i = \sum_{j,k} \Gamma_{ijk} \dot{q}_j \dot{q}_k, \qquad \Gamma_{ijk} = \tfrac{1}{2}(\partial_{q_j} M_{ik} + \partial_{q_k} M_{ij} - \partial_{q_i} M_{jk}). \tag{S12}$$

Writing $p = M(q)\dot{q}$ gives

$$\dot{p}_i = -\tfrac{1}{2} p^{\mathsf{T}} (\partial_{q_i} M^{-1}) p - \partial_{q_i} V - \partial_{v_i} \mathcal{D}_N + (Q_{\mathrm{drv}} + B_u u)_i. \tag{S13}$$

If the dissipation is stored as a momentum readout $G_1(q,p) = \mathcal{D}_N(q, M^{-1}p)$, its physical force is

$$-\partial_v \mathcal{D}_N = -M(q) \nabla_p G_1, \tag{S14}$$

not generally $-\nabla_p G_1$. The Christoffel term already follows from H_q and is not placed in G_1. The energy identity is

$$\dot{H} = -v^{\mathsf{T}} \partial_v \mathcal{D}_N + v^{\mathsf{T}}(Q_{\mathrm{drv}} + B_u u). \tag{S15}$$

Explicit time dependence adds $\partial_t H$; a time-dependent mass also generates the corresponding inertial terms before the Legendre transform. A quadratic positive semidefinite Rayleigh potential, or a differentiable convex velocity potential minimized at zero, ensures $v^{\mathsf{T}} \partial_v \mathcal{D}_N \geq 0$. Nonnegativity of the potential alone would not ensure that inequality.

S1.2 Weak form, endpoint inertia and consistent Hermite matrices

For an admissible test displacement η with $\eta(0) = \eta_s(0) = 0$, the benchmark weak form is

$$\int_0^L \rho A_c w_{tt} \eta \, \mathrm{d}s + m_t w_{tt}(L)\eta(L) + J_t \ddot{\vartheta} \eta_s(L) + \alpha \left[\int_0^L \rho A_c w_t \eta \, \mathrm{d}s + m_t w_t(L)\eta(L) + J_t \dot{\vartheta} \eta_s(L) \right] + \int_0^L EI w_{ss} \eta_{ss} \, \mathrm{d}s = F\eta(L) + [\tau_B - \mu B_0 \sin\vartheta]\eta_s(L). \tag{S16}$$

Two spatial integrations by parts give the interior and endpoint equations in (5)–(6). The J_t term is a tip rigid-body inertia, distinct from the distributed rotary inertia omitted by the Euler–Bernoulli model.

For an element of length ℓ_e and local coordinate $\xi \in [0,1]$, the displacement row is

$$N_e(\xi)^{\mathsf{T}} = \begin{pmatrix} 1-3\xi^2+2\xi^3 \\ \ell_e(\xi-2\xi^2+\xi^3) \\ 3\xi^2-2\xi^3 \\ \ell_e(-\xi^2+\xi^3) \end{pmatrix}. \tag{S17}$$

The local order is $(w_i, \vartheta_i, w_{i+1}, \vartheta_{i+1})$. Integrating the bending and kinetic energies yields

$$K_e = \frac{EI}{\ell_e^3}\begin{pmatrix} 12 & 6\ell_e & -12 & 6\ell_e \\ 6\ell_e & 4\ell_e^2 & -6\ell_e & 2\ell_e^2 \\ -12 & -6\ell_e & 12 & -6\ell_e \\ 6\ell_e & 2\ell_e^2 & -6\ell_e & 4\ell_e^2 \end{pmatrix},$$
$$M_e = \frac{\rho A_c \ell_e}{420}\begin{pmatrix} 156 & 22\ell_e & 54 & -13\ell_e \\ 22\ell_e & 4\ell_e^2 & 13\ell_e & -3\ell_e^2 \\ 54 & 13\ell_e & 156 & -22\ell_e \\ -13\ell_e & -3\ell_e^2 & -22\ell_e & 4\ell_e^2 \end{pmatrix}. \tag{S18}$$

After assembly and elimination of the two clamped degrees of freedom, add m_t to the terminal displacement mass entry and J_t to the terminal rotation entry. Add μB_0 to the terminal rotation stiffness entry only when forming the biased linear eigenproblem. The nonlinear physical energy still uses K_{beam} and the cosine potential.

For an axial centre-of-mass offset e from the endpoint, the rigid-body kinetic contribution would instead contain the block

$$\begin{pmatrix} m_t & m_t e \\ m_t e & J_{\mathrm{cm}} + m_t e^2 \end{pmatrix} \tag{S19}$$

in endpoint translation and rotation coordinates. The benchmark sets $e = 0$. Coating, adhesive and mounting masses are not represented by this ideal bare-core inertia.

The nodal displacement entries have units of length and the slope entries are dimensionless. Consequently, the (w,w), (w,ϑ) and (ϑ,ϑ) blocks of M_f have units kg, kg m and kg m^2, respectively. The corresponding stiffness units are N/m, N and N m; the generalized force entries are N and N m. These mixed units explain why neither a scalar mass replacement nor an unscaled Euclidean actuator condition number is physically neutral.

S1.3 Linear frequency verification and parameter sensitivity

An independent continuous-beam check is available because the biased linearized endpoint problem has constant coefficients. Put

$$\lambda = kL, \qquad \omega^2 = \frac{EI}{\rho A_c L^4}\lambda^4,$$
$$w(s) = A[\cosh(ks) - \cos(ks)] + B[\sinh(ks) - \sin(ks)]. \tag{S20}$$

The undamped endpoint conditions for this eigenproblem are

$$EIw_{sss}(L) + m_t\omega^2 w(L) = 0,$$
$$EIw_{ss}(L) + (\mu B_0 - J_t\omega^2)w_s(L) = 0. \tag{S21}$$

Define $\eta_t = m_t/(\rho A_c L)$, $\iota_t = J_t/(\rho A_c L^3)$, $\kappa_t = \mu B_0 L/(EI)$ and

$$f = \cosh\lambda - \cos\lambda, \qquad g_\lambda = \sinh\lambda - \sin\lambda,$$
$$c_+ = \cosh\lambda + \cos\lambda, \qquad s_+ = \sinh\lambda + \sin\lambda.$$

Writing $h_\lambda = \kappa_t/\lambda - \iota_t\lambda^3$, the positive roots satisfy

$$\det\begin{pmatrix} g_\lambda + \eta_t\lambda f & c_+ + \eta_t\lambda g_\lambda \\ c_+ + h_\lambda s_+ & s_+ + h_\lambda f \end{pmatrix} = 0. \tag{S22}$$

For zero tip inertia and zero bias this reduces to $1+\cosh\lambda\cos\lambda = 0$. The damped frequencies are not used in mass normalization; ω_j throughout the paper denotes the undamped frequency of the biased stiffness.

The eigenvectors are mass-normalized and oriented so that each tip displacement coefficient is positive. The 32-element maximum mass-orthonormality defect is 5.14×10^{-15}; the first two relative frequency errors against (S22) are 9.02×10^{-9} and 3.29×10^{-7}. At fixed bias, omitting tip inertia gives $(\omega_1,\omega_2) = (7.859186509, 47.517207103)$ rad/s. Including it lowers the frequencies by approximately 0.2440% and 0.2476%. The linear verification does not bound errors caused by neglecting nonlinear high-mode response.

Table S2 varies parameters one at a time. The B_r endpoints come from the N50 grade interval (SM Magnetics 2026). The inertia and modulus perturbations are declared design studies, not measured confidence intervals. Inertia perturbations scale m_t and J_t together at fixed core dimension and fixed magnetic moment. For reported damping-ratio changes in the machine-readable record, the nominal α is held fixed.

The combined B_r and $\pm10\%$ inertia corners change ω_1 by between -0.10729% and $+0.10705\%$. The tested modulus changes therefore dominate those magnetic-core perturbations in this frequency comparison. They do not establish a bound on all possible fabrication variation.

The material and density reference is a particular Sylgard preparation (Cho et al. 2021). Mechanical loss

Table S1. Angular frequencies in rad/s for the revised beam, including magnetic-core translational and rotational inertia. The final row solves the independent continuous-beam endpoint problem.

Discretization	ω_1	ω_2	ω_3	ω_4
16 Hermite elements	7.840010222	47.399809441	131.211581398	256.375395140
32 Hermite elements	7.840009162	47.399576624	131.206681073	256.339221935
64 Hermite elements	7.840009096	47.399561992	131.206371332	256.336917925
Continuous endpoint equation	7.840009091	47.399561016	131.206350623	256.336763590

Table S2. Design sensitivity of the first two biased frequencies. Percentages are relative to the nominal 32-element model.

Change	$\Delta\omega_1/\omega_1$	$\Delta\omega_2/\omega_2$	Interpretation
$B_r = 1.40$ T	-0.082935%	-0.029460%	Grade-range endpoint
$B_r = 1.45$ T	$+0.082656\%$	$+0.029413\%$	Grade-range endpoint
m_t, J_t reduced by 10%	$+0.024384\%$	$+0.024646\%$	Chosen design perturbation
m_t, J_t increased by 10%	-0.024367%	-0.024608%	Chosen design perturbation
E reduced by 20%	-9.526743%	-10.186350%	Chosen design perturbation
E increased by 20%	$+8.668840\%$	$+9.235817\%$	Chosen design perturbation

depends on preparation and measurement conditions; loss-factor measurements for other samples (Fischer et al. 2017) are not converted into the present modal damping ratios without a constitutive and frequency correspondence. The scenarios $\zeta_1^{\mathrm{d}} \in \{0, 0.01, 0.025, 0.05, 0.1\}$ instead define explicit effective damping models. Likewise, the core dimensions are design inputs and the remanent-field midpoint is nominal; the available specifications do not calibrate a coated, bonded specimen.

S1.4 Canonical units, complex derivatives and input conversion

After mass normalization, a_j has units $\sqrt{\mathrm{kg}}\,\mathrm{m}$ and $\pi_j = \dot{a}_j$ has units $\sqrt{\mathrm{kg}}\,\mathrm{m\,s^{-1}}$. Let $P_s = \mathrm{diag}(\sqrt{E_0\omega_j/\omega_0})$. The scales in (11) satisfy

$$D_s P_s = (E_0/\omega_0) I,$$
$$\sum_j \mathrm{d}a_j \wedge \mathrm{d}\pi_j = \frac{E_0}{\omega_0} \sum_j \mathrm{d}X_j \wedge \mathrm{d}Y_j. \tag{S23}$$

With $h = H/E_0$ and $\tau = \omega_0 t$, the chain rule gives $X' = h_Y$, $Y' = -h_X$. The physical quadratic kinetic energy becomes $E_0 Y^{\mathsf{T}} \mathsf{N} Y/2$, while the Lagrangian kinetic energy in dimensionless position and velocity is $v^{\mathsf{T}} \mathsf{N}^{-1} v/2$. The three distinct masses are therefore the nodal M_f, the mass-normalized modal identity and the balanced $M_{\mathrm{eff}} = \mathsf{N}^{-1}$.

The physical Rayleigh potential $\mathcal{D}_{\mathrm{phys}} = \alpha \dot{a}^{\mathsf{T}} \dot{a}/2$ transforms as

$$\mathcal{D}(X, v) = \frac{\mathcal{D}_{\mathrm{phys}}(D_s X, \omega_0 D_s v)}{E_0 \omega_0} = \tfrac{1}{2} \gamma v^{\mathsf{T}} \mathsf{N}^{-1} v. \tag{S24}$$

For the fixed kinetic energy, $v = \mathsf{N} Y$ and its force is $-\partial_v \mathcal{D} = -\gamma Y$. If learned momentum terms change h_Y, the force remains $-\gamma \mathsf{N}^{-1} h_Y$, rather than automatically $-\gamma Y$. The realization in (S36) handles a general target through its actual velocity map.

For $\zeta = (X + \mathrm{i}Y)/\sqrt{2}$ and $\chi = (X - \mathrm{i}Y)/\sqrt{2}$, the inverse chart gives

$$\partial_\chi h^{\mathbb{C}} = (\partial_X h + \mathrm{i}\partial_Y h)/\sqrt{2}. \tag{S25}$$

Substituting the canonical equations into ζ' proves (17) with no additional factor of two. For a force Q acting in the Y equation, the extra term in ζ' is $\mathrm{i}Q/\sqrt{2}$. A proposed complex readout $h + \mathrm{i}\mathcal{D}$ must therefore be used with the decoder (18); differentiating that sum as though it were one conservative Hamiltonian generally produces another vector field.

For two modes, b_j has units $\mathrm{kg}^{-1/2}$ and t_j has units $\mathrm{kg}^{-1/2}\mathrm{m}^{-1}$. Both $b_j F$ and $t_j \tau_B$ consequently have the units of modal acceleration. From $X' = \mathsf{N} Y$,

$$\ddot{a} = \omega_0^2 D_s \mathsf{N} Y' = \Lambda Y', \qquad \Lambda = E_0 D_s^{-1}. \tag{S26}$$

Moreover, $d = D_s \boldsymbol{t}$ gives

$$\Lambda \nabla V_0 = \Omega_2^2 a + \boldsymbol{t} E_0 (\sin\vartheta - \vartheta). \tag{S27}$$

Equations (S26)–(S27) prove (73). If a program stores the total modal moment T, it must first remove $E_0(\vartheta - \sin\vartheta)$ before dividing by $\mu\cos\vartheta$ to obtain B_y. This subtraction changes a port convention; it does not compensate the unknown residual $-\Lambda\nabla R_*$. For the learned input, $\Lambda\widehat{\gamma} Y = \omega_0 \widehat{\gamma}\dot{a} = \widehat{\alpha}\dot{a}$. The physical potential forces combine as $-\Lambda\nabla R_* + \Lambda\nabla\widehat{R} = -\Lambda\nabla\delta R$, which completes (75). No true residual derivative is used by the fitted controller; it is evaluated only by the synthetic plant or its explicitly labeled truth-only comparator.

The numerical conversion uses

$$\begin{aligned} A_{\mathrm{mag}} &\simeq \begin{pmatrix} 203.96232682 & 3303.68875398 \\ 205.00706618 & 11909.57162007 \end{pmatrix}, \\ D_s &= \mathrm{diag}(S_1, S_2), \\ S_1 &\simeq 9.29943515 \times 10^{-5}, \\ S_2 &\simeq 3.78205312 \times 10^{-5}. \end{aligned} \tag{S28}$$

Its determinant is nonzero. Rescaling the columns by actual available force and torque is needed to assess a device's effective authority; column normalization alone verifies only geometric separation of the two generalized directions.

S1.5 Residual compensation under modal enrichment

The full retained-mode potential corresponding to (79) is

$$\begin{aligned} V_N = \tfrac{1}{2} a^{(N)\mathsf{T}} \Omega_N^2 a^{(N)} + E_0 g(\vartheta_N) \\ + \mathcal{R}_N(a^{(N)}). \end{aligned} \tag{S29}$$

Its kinetic energy is $\dot{a}^{(N)\mathsf{T}} \dot{a}^{(N)}/2$. The quadratic magnetic bias is already in Ω_N^2, so the same subtraction in g avoids double counting. All potentials $\mathcal{R}_N$ in a comparison are restrictions of one specified full potential. The compatibility condition in the main text ensures that setting the high coordinates to zero recovers the original synthetic residual, without asserting that its continuation was identified experimentally.

For the first two modal equations,

$$\begin{aligned} \Lambda^{-1} \Omega_2^2 a_{1:2} &= \mathsf{N} X, \\ \Lambda^{-1} E_0 \boldsymbol{t}_{1:2} &= d, \\ \Lambda^{-1} \nabla_{a_{1:2}} \mathcal{R}_N &= \nabla R_*(X) + e_R^{(N)}. \end{aligned} \tag{S30}$$

Dividing their acceleration equation by Λ and inserting $U_{\mathrm{rec}} = \widehat{\gamma} Y + \nabla \widehat{R} + e_{\mathrm{port}}$ proves (81). In particular, $\vartheta_N - \vartheta_2 = \sum_{j=3}^{N} t_j a_j$. If both angles lie in $[-\vartheta_{\max}, \vartheta_{\max}]$, the mean-value theorem and $1 - \cos s \le s^2/2$ give

$$\|e_{\mathrm{mag}}^{(N)}\| \le \tfrac{1}{2} \|d\| \vartheta_{\max}^2 |\vartheta_N - \vartheta_2|. \tag{S31}$$

A bound on high-mode motion is still needed to use this inequality. Sampling small angles on a two-mode trajectory does not supply it.

For the declared residual $R_*(X) = \beta_* g(\vartheta_2)$, one concrete synthetic extension for the planned modal comparison is

$$\mathcal{R}_N^{\mathrm{ang}}(a^{(N)}) = E_0 \beta_* g(\vartheta_N). \tag{S32}$$

It preserves the low-dimensional restriction and gives $e_R^{(N)} = -\beta_* e_{\mathrm{mag}}^{(N)}$. Thus the combined magnetic and residual difference is $(1 + \beta_*) e_{\mathrm{mag}}^{(N)}$, even with perfect coefficient identification. As an isolating comparison, the alternative $\mathcal{R}_N^{\mathrm{low}} = E_0 R_*(D_s^{-1} a_{1:2})$ gives $e_R^{(N)} = 0$ and no residual force in the higher coordinates. These are different declared synthetic potentials, not equivalent consequences of the same two-mode observations. Neither higher-mode comparison has been executed in the present numerical record.

Let $H_{0,N}$ omit $\mathcal{R}_N$ from the full retained energy. For the applied physical ports,

$$\begin{aligned} \dot{H}_{0,N} = -\alpha \|\dot{a}^{(N)}\|^2 - \dot{a}^{(N)\mathsf{T}} \nabla_a \mathcal{R}_N \\ + F_{\mathrm{app}} \dot{w}(L) + \tau_{B,\mathrm{app}} \dot{\vartheta}_N. \end{aligned} \tag{S33}$$

The power contributed by the residual must be included in a nominal-energy work bound. Alternatively, the augmented energy may be used after a suitable lower bound by bending energy has been established. Its nonnegative decomposition cannot be assumed from the two-mode residual ansatz.

S1.6 A stronger exact realization of a conservative target

This construction uses known damping and exact target derivatives. It is retained as a conditional full-model result separate from the unchanged fitted feedback in (81). For the nominal finite-element plant (7), retain all $N_f = 64$ modes and define

$$\begin{aligned} \ddot{a}_j + \alpha \dot{a}_j + \omega_j^2 a_j = b_j F + t_j T, \qquad j = 1, \dots, N_f, \\ T = E_0(\vartheta - \sin\vartheta) + \tau_B, \\ \vartheta = \sum_{j=1}^{N_f} t_j a_j. \end{aligned} \tag{S34}$$

The symbol T is a total modal moment, whereas U remains the dimensionless generalized input. Equation (S34) specifies the full plant for the following result; an arbitrary unmodelled force on omitted modes is not included in its hypotheses.

Consider a real analytic target Hamiltonian $h_{\mathrm{tar}}(X, Y)$, which may contain momentum-dependent terms. With physical low-mode velocity $v_{1:2} = \dot{a}_{1:2}$, infer Y from

$$(h_{\mathrm{tar}})_Y(X, Y) = \omega_0^{-1} D_s^{-1} v_{1:2}, \qquad X = D_s^{-1} a_{1:2}. \tag{S35}$$

On a domain with positive definite $(h_{\mathrm{tar}})_{YY}$ and convex Y-fibres this relation is injective in Y and locally invertible. The acceleration and port requests that reproduce the canonical

target are

$$
\begin{aligned}
a^{\mathrm{des}}_{1:2,tt} &= D_s\omega_0^2\big[(h_{\mathrm{tar}})_{YX}(h_{\mathrm{tar}})_Y - (h_{\mathrm{tar}})_{YY}(h_{\mathrm{tar}})_X\big],\\
\begin{pmatrix} F \\ T \end{pmatrix} &= A_{\mathrm{mag}}^{-1}\big[a^{\mathrm{des}}_{1:2,tt} + \alpha v_{1:2} + \Omega_2^2 a_{1:2}\big],\\
B_y &= \frac{T - E_0(\vartheta - \sin\vartheta)}{\mu\cos\vartheta}.
\end{aligned}
\tag{S36}
$$

The actual full tip angle enters the last line. Substitution first enforces the desired low acceleration; differentiating (S35) then recovers both canonical equations.

Proposition S1. Conditional finite-element lift. *Suppose h_{tar} has an analytic invariant embedded two-torus carrying a quasiperiodic flow, and the velocity map in (S35) satisfies the stated positivity and fibre conditions on a neighbourhood of it. Assume $\alpha > 0$, A_{mag} is invertible and the exact feedback (S36) is available. If the resulting response graph lies inside the allowed angle and actuator domain, it is an invariant two-torus of (S34). For each fixed low-mode orbit on this torus, higher-mode deviations decay exponentially while the closed-loop trajectory remains in that domain.*

The feedback in (S36) cancels the nominal full-angle remainder to impose a chosen low acceleration. It does not contain an arbitrary unknown full residual $\mathcal{R}_N$, and it is not obtained merely by applying the learned two-mode law to more modes. The result concerns a finite-dimensional plant; no infinite-dimensional beam limit or attraction of arbitrary low-mode initial conditions is asserted. The proof and its domain requirements follow below.

S1.7 Proof and scope of the finite-element lift

The proof uses the nominal full modal plant (S34), known parameters and exact feedback. Let $\iota(\phi) = (X(\phi), Y(\phi))$ be the target torus, with $\phi \in (\mathbb{R}/2\pi\mathbb{Z})^2$ and physical frequency Ω. Its physical low-mode image is

$$
a_{1:2} = D_s X(\phi), \qquad v_{1:2} = \omega_0 D_s (h_{\mathrm{tar}})_Y(\iota(\phi)). \tag{S37}
$$

If two points in this image have the same position, they have the same X. For distinct Y_1, Y_2 in the same convex fibre, write $\Delta Y = Y_1 - Y_2$, $Y_s = Y_2 + s\Delta Y$ and $h = h_{\mathrm{tar}}$. Then

$$
\begin{aligned}
&\Delta Y^{\mathsf{T}}[h_Y(X, Y_1) - h_Y(X, Y_2)]\\
&\quad = \int_0^1 \Delta Y^{\mathsf{T}} h_{YY}(X, Y_s)\Delta Y \,\mathrm{d}s > 0.
\end{aligned}
\tag{S38}
$$

Thus equality of the physical velocity also implies $Y_1 = Y_2$. The differential of the velocity map is invertible in the Y directions, so it preserves the embedded low torus locally and globally on the stated domain.

The requested F, T in (S36) depend only on the low physical state. The full tip angle enters solely in conversion from T to B_y, which cancels the same nonlinear bias remainder in every modal equation. Each higher mode then has prescribed quasiperiodic forcing. For a real temporal frequency σ,

$$
D_j(\sigma) = \omega_j^2 - \sigma^2 + \mathrm{i}\alpha\sigma \tag{S39}
$$

cannot vanish: at $\sigma = 0$ its real part is positive, and at nonzero σ its imaginary part is nonzero. Its modulus tends to infinity as $|\sigma| \to \infty$, so its reciprocal is bounded on the real axis. Using the Fourier convention $e^{\mathrm{i}k\cdot\phi}$, the unique bounded higher-mode response therefore has coefficients

$$
a^*_{j,k} = \frac{b_j F_k + t_j T_k}{\omega_j^2 - (k\cdot\Omega)^2 + \mathrm{i}\alpha(k\cdot\Omega)}, \qquad j \geq 3. \tag{S40}
$$

Consequently the exponentially decaying Fourier coefficients of F, T remain exponentially decaying after multiplication by $D_j(k\cdot\Omega)^{-1}$. This establishes an analytic response for each of the finitely many higher modes.

Differences from that response satisfy $\ddot{z}_j + \alpha\dot{z}_j + \omega_j^2 z_j = 0$ and decay exponentially. The low motion remains neutral along its target torus; no attraction of general low initial conditions follows. With $\alpha = 0$, the resolvent may have resonances and this particular lifting argument does not apply. That distinction includes the zero-damping sensitivity scenario.

The result requires the complete response graph and any considered startup transient to remain in the angle, sensing and actuator domain. It does not supply those bounds merely from the two-mode fit. In particular, the reduced synthetic potential residual is not a specified full-space model of omitted-mode forces. Extending the theorem to such forces requires an explicit full model and its corresponding estimates. Similarly, a governor that changes F, T breaks exact target realization even when it preserves clearance.

S1.8 Whole-centreline energy bound and initialization

For $w \in H^2(0, L)$ with $w(0) = w_s(0) = 0$,

$$
\begin{aligned}
|w(s)|^2 &\leq \left(\int_0^s (s-\xi)^2\,\mathrm{d}\xi\right)\left(\int_0^s |w_{ss}|^2\,\mathrm{d}\xi\right)\\
&\leq \frac{2s^3}{3EI}E_{\mathrm{bend}}, \qquad E_{\mathrm{bend}} = \frac{EI}{2}\int_0^L |w_{ss}|^2\,\mathrm{d}s.
\end{aligned}
\tag{S41}
$$

Since the kinetic energy includes a positive tip mass and inertia, and $E_0(1-\cos\vartheta) \geq 0$, the nominal total energy

satisfies $E \geq E_{\mathrm{bend}}$. Taking the supremum proves (83). This also holds for conforming C^1 Hermite displacement fields. The related slope estimate is

$$\|w_s\|_\infty^2 \leq L \int_0^L |w_{ss}|^2 \,\mathrm{d}s \leq \frac{2LE}{EI}. \tag{S42}$$

These inequalities require a bound on the relevant total physical energy, not just on one modal amplitude or the tip displacement.

The constant in (S41) is sharp for the bending energy at the tip: equality is attained by a curvature proportional to $L - s$, the static tip-force shape. That equality case explains the familiar expression $E_c = 3EIc^2/(2L^3)$ but does not restrict the inequality to static shapes. For the revised beam $E_c = 10.0197171$ nJ. An energy cap $E_{\max} = 0.95E_c = 9.5187312$ nJ would imply $\|w\|_\infty \leq \sqrt{0.95}\, c < c$.

An initialization bound can use the full-state quadratic form

$$Q(q, v) = \tfrac{1}{2} v^{\mathsf{T}} M_f v + \tfrac{1}{2} q^{\mathsf{T}} K_b q. \tag{S43}$$

Since $1 - \cos\vartheta \leq \vartheta^2/2$, $E(q, v) \leq Q(q, v)$. If estimates $(\widehat{q}, \widehat{v})$ satisfy $Q(q - \widehat{q}, v - \widehat{v}) \leq \mathcal{E}_{\mathrm{init}}$, then

$$E(q, v) \leq \left[\sqrt{Q(\widehat{q}, \widehat{v})} + \sqrt{\mathcal{E}_{\mathrm{init}}}\right]^2. \tag{S44}$$

This follows from the triangle inequality for $\sqrt{Q}$. Tip-only measurements do not by themselves bound energy in unobserved bending or velocity components.

For the stored nominal two-mode initial state $(X, Y) = (0.09, 0.08, 0.015, -0.01)$, direct substitution into $E_0 h_0$ gives 12.6567207 nJ. This exceeds $E_{\max}$, so the work-governor initialization condition has not been verified for the reported compensation runs. Their sampled centreline values can be below the comparison clearance without satisfying this sufficient energy condition. The tracking initialization belongs to the separate extension in Supplementary Section S4.

For a plant augmented by R_*, the nominal-energy derivative gains the power of its additional generalized force. Alternatively, one could use the augmented energy only after proving a suitable lower bound by bending energy. The synthetic residual alone is therefore not silently incorporated into the nominal clearance theorem.

S1.9 *Input-work governor and bounded implementation errors*

For the nominal power identity and exact port measurements,

$$E(t) - W(t) = E(0) - W(0) - \int_0^t \alpha \dot{q}^{\mathsf{T}} M_f \dot{q} \,\mathrm{d}s \leq 0. \tag{S45}$$

The constraint in (85) implies $\dot{W} \leq \kappa(E_{\max} - W)$. With $W(0) < E_{\max}$, comparison gives $E_{\max} - W(t) \geq [E_{\max} - W(0)]e^{-\kappa t}$. The governor has a feasible zero active input whenever the prescribed bias itself is feasible. For a nonempty closed convex admissible port set containing zero and a positive definite scaled cost, its quadratic projection is unique.

The same construction can accommodate deterministic error envelopes, provided they are supplied independently. Let the applied ports obey

$$\begin{aligned} F &= (1 + g_F)u_F + e_F, \\ B_y &= (1 + g_B)u_B + e_B, \\ |g_i| &\leq \eta_i, \qquad |e_i| \leq \epsilon_i. \end{aligned} \tag{S46}$$

For errors bounded by $|v_w - \widehat{v}_w| \leq \delta_v$, $|\vartheta - \widehat{\vartheta}| \leq \delta_\vartheta$ and $|\dot{\vartheta} - \widehat{\dot{\vartheta}}| \leq \delta_\omega$, define

$$\begin{aligned} \widehat{a}_F &= \widehat{v}_w, & \Delta_F &= \delta_v, \\ \widehat{a}_B &= \mu \cos\widehat{\vartheta}\,\widehat{\dot{\vartheta}}, & \Delta_B &= \mu(\delta_\omega + |\widehat{\dot{\vartheta}}|\delta_\vartheta). \end{aligned} \tag{S47}$$

The cosine is one-Lipschitz, so $|a_i - \widehat{a}_i| \leq \Delta_i$. A convex positively homogeneous upper bound on the non-additive port power is

$$P^+(u) = \sum_{i \in \{F,B\}} \left[\widehat{a}_i u_i + \{\Delta_i + \eta_i(|\widehat{a}_i| + \Delta_i)\}|u_i|\right]. \tag{S48}$$

It contains no division by a measured velocity and satisfies $P^+(0) = 0$.

For $\alpha > 0$, define inverse-mass coefficients $m_F = e_w^{\mathsf{T}} M_f^{-1} e_w$ and $m_\vartheta = e_\vartheta^{\mathsf{T}} M_f^{-1} e_\vartheta$. The additive generalized force $r_e = e_w e_F + \mu e_\vartheta e_B \cos\vartheta$ obeys

$$\begin{aligned} r_e^{\mathsf{T}} v - \alpha v^{\mathsf{T}} M_f v \leq \frac{r_e^{\mathsf{T}} M_f^{-1} r_e}{4\alpha} \leq p_e, \\ p_e = \frac{(\epsilon_F \sqrt{m_F} + \mu \epsilon_B \sqrt{m_\vartheta})^2}{4\alpha}. \end{aligned} \tag{S49}$$

Completion of the square proves the first inequality, and the triangle inequality in the M_f^{-1} norm proves the second. Thus $\dot{E} \leq P^+(u) + p_e$.

On a chosen horizon $[0, T_H]$, reserve $R_H = T_H p_e$ and put $C_H = E_{\max} - R_H$. If $W(0) \geq E(0)$ and $W(0) < C_H$, integrate $\dot{W} = P^+(u)$ and impose

$$P^+(u) \leq \kappa(C_H - W), \qquad u \in \mathcal{U}(q). \tag{S50}$$

Then, throughout $0 \leq t \leq T_H$,

$$E(t) \leq W(t) + t p_e < C_H + R_H = E_{\max}. \tag{S51}$$

The command set must reserve the stated gain and additive errors when enforcing actual actuator limits. If additive errors vanish, the reserve is zero and the exact-power argument extends to all time for $\alpha \geq 0$. A nonzero constant additive-error allowance gives the stated finite-horizon condition, not an all-time conclusion.

For sample-and-hold execution, a bound at one sampling instant is insufficient. If $P_k^+(u_k)$ bounds power throughout the hold interval of length h_k, the rule

$$h_k P_k^+(u_k) \leq C_H - W_k, \qquad W_{k+1} = W_k + h_k P_k^+(u_k) \tag{S52}$$

also bounds the energy within that interval. The error envelope must cover sensor age, hold age and command latency. For example, a known command slew bound $|\dot{u}_i| \leq S_i$ and delay d_i may be included by increasing ϵ_i by $(1+\eta_i)S_i d_i$. These are conditional implementation bounds; no numerical sensor, latency or coil-current envelope has been calibrated for the revised benchmark. The reported runs test the ideal physical input conversion, while this work argument states what an energy-certified implementation would additionally have to satisfy.

S2 Geometric identities and learning-to-KAM conditions

S2.1 Compatibility of the Birkhoff representation

Let $\alpha_t = \mathcal{R}(z,t)^{\mathsf{T}}\,\mathrm{d}z$ and let the coefficient matrix of $\mathrm{d}\alpha_t$ be $\mathcal{K}_{ij} = \partial_i \mathcal{R}_j - \partial_j \mathcal{R}_i$. With this convention, the canonical one-form $p^{\mathsf{T}}\,\mathrm{d}q$ has $\mathcal{K} = -J$. If $b = \mathcal{K}\dot{z}$, the representation (21) implies

$$\begin{aligned} \partial_k \mathcal{K}_{ij} + \partial_i \mathcal{K}_{jk} + \partial_j \mathcal{K}_{ki} &= 0, \\ \partial_t \mathcal{K}_{ij} &= \partial_i b_j - \partial_j b_i. \end{aligned} \tag{S53}$$

These identities, together with $\det \mathcal{K} \neq 0$, explain why a dissipative force cannot be inserted into a fixed canonical two-form by freely assigning a scalar Birkhoff function (Su et al. 2007; Feng and Qin 2010).

For a sufficiently smooth local flow Φ_{t,t_0} of a general vector field $v(z,t)$, a local existence construction is available. Choose an initial one-form α_0 with nondegenerate $\mathrm{d}\alpha_0$ and transport it by the inverse flow:

$$\alpha_t = (\Phi_{t,t_0}^{-1})^* \alpha_0, \qquad \mathcal{B} = \alpha_t(v).$$

The transport equation $\partial_t \alpha_t + \mathcal{L}_v \alpha_t = 0$ and Cartan's identity give $\mathcal{K} v = \mathrm{d}\mathcal{B} + \partial_t \alpha_t$. This is a local, flow-dependent representation, not a computable Hamiltonian reduction supplied by the mechanical coefficients alone. It neither establishes a global analytic chart nor an autonomous integrable reference.

The proportional-damping choice in (23) is constructive. Indeed,

$$-aJ\begin{pmatrix}\dot{q}\\ \dot{p}\end{pmatrix} = \begin{pmatrix} aH_q + \dot{a}\,p \\ aH_p \end{pmatrix},$$

so $\dot{q} = H_p$ and $\dot{p} = -H_q - \alpha p$. The substitution $\Pi = ap$ gives $\dot{q} = \widetilde{H}_\Pi$ and $\dot{\Pi} = -\widetilde{H}_q$ even for $M(q)$. For constant mass, applying the time-midpoint rule to $\widetilde{H} = e^{-\alpha t}\Pi^{\mathsf{T}} M^{-1}\Pi/2 + e^{\alpha t} V(q)$ and using $e^{-\alpha t_m}\bar{\Pi} = (r p_+ + r^{-1} p_-)/2$ proves (27). The coordinate map $T_t(q,p) = (q, e^{\alpha t} p)$ satisfies $DT_t^{\mathsf{T}} J DT_t = e^{\alpha t} J$. Symplecticity in (q,Π) therefore pulls back to

$$D\Phi^{\mathsf{T}} \mathcal{K}(t_+) D\Phi = \mathcal{K}(t_-), \qquad D\Phi^{\mathsf{T}} J D\Phi = e^{-\alpha \Delta_t} J.$$

Explicit time dependence must be evaluated at t_m for the stated second-order method. For variable mass the complete $\widetilde{H}_q$ is retained, rather than reusing the constant-mass update.

S2.2 Endpoint projection, dimensions and the implicit branch

Write $Z = (q,\Pi) \in \mathbb{R}^{2n}$ and introduce

$$T_0 = \mathrm{diag}(I_n, -I_n), \qquad \mathsf{E} = \begin{pmatrix} 0 & I_n \\ I_n & 0 \end{pmatrix}.$$

The endpoint transformation is $\widehat{w} = \mathsf{E}(Z_+ - Z_-)$, $w = T_0(Z_+ + Z_-)/2$; thus

$$D\alpha = \begin{pmatrix} \mathsf{E} & -\mathsf{E} \\ T_0/2 & T_0/2 \end{pmatrix} \in \mathbb{R}^{4n\times 4n}. \tag{S54}$$

The compatibility identity can be checked without assuming a generating function in advance:

$$\widehat{w}^{\mathsf{T}}\,\mathrm{d}w = (\Pi_+ - \Pi_-)^{\mathsf{T}}\,\mathrm{d}\bar{q} - (q_+ - q_-)^{\mathsf{T}}\,\mathrm{d}\bar{\Pi}.$$

Taking its exterior derivative gives $\mathrm{d}\lambda_{Z_+} - \mathrm{d}\lambda_{Z_-}$, with $\lambda_Z = \Pi^{\mathsf{T}}\,\mathrm{d}q$. Thus it is closed on a symplectic graph between fixed time endpoints. Once the projection onto w is locally invertible, the Poincaré lemma supplies the scalar generator in Proposition 1. This verifies both compatibility and the particular projection used by the construction.

On a flow or discrete graph $Z_+ = \Phi(Z_-)$,

$$D_{Z_-} w = \tfrac{1}{2} T_0 (D\Phi + I_{2n}). \tag{S55}$$

At zero step this is T_0, so it is locally invertible for sufficiently small steps. The phase-flow Jacobian $D\Phi$ is $2n \times 2n$ and is unrelated to the size of the mechanical mass matrix. The original endpoint variables are recovered jointly from $q_\pm = Q \pm \widehat{P}/2$ and $\Pi_\pm = -P \pm \widehat{Q}/2$. Invertibility of the graph function $\widehat{w} = f(w)$ is not required: at zero step it is identically zero and $f_w = 0$.

In the endpoint convention, the time-midpoint generator is $\varphi_{\Delta_t}(Q,P) = -\Delta_t \widetilde{H}(Q,-P,t_m)$. Because $P = -\bar{\Pi}$, its two derivatives give $\Pi_+ - \Pi_- = -\Delta_t \widetilde{H}_q$ and $q_+ - q_- = \Delta_t \widetilde{H}_\Pi$ with the correct physical signs. The autonomous convention S_Δ uses $\varphi_\Delta(Q,P) = -S_\Delta(Q,-P)$. The generator therefore changes together with the lifted momentum; the physical p and lifted Π cannot be interchanged in only part of the derivation.

For the central method, let $\mathcal{F}(x_+;x_-) = x_+ - x_- - J\nabla S_\Delta((x_+ + x_-)/2)$. Its Newton matrix and exact map derivative are

$$\begin{aligned} &\partial_{x_+}\mathcal{F} = I_{2n} - C_\Delta, \qquad C_\Delta = \tfrac{1}{2} J S''_\Delta(m), \\ &D\Psi_\Delta = (I_{2n} - C_\Delta)^{-1}(I_{2n} + C_\Delta). \end{aligned} \tag{S56}$$

Invertibility of $I - C_\Delta$ gives a locally unique branch by the implicit function theorem, and analyticity of the data gives analyticity of that branch. Equation (S55) then becomes $D_{x_-} w = T_0 (I - C_\Delta)^{-1}$. The endpoint projection and implicit solve therefore share the same local nondegeneracy condition. Implementations solve the Newton linear system; no explicit matrix inverse is needed.

S2.3 Noncommuting-matrix elimination in ordinary midpoint

For constant matrices, the physical equations are $\dot{q} = M^{-1}p$ and $\dot{p} = -Kq - CM^{-1}p + F + U$, with M positive definite and K symmetric. Ordinary midpoint gives

$$\bar{q} = q_- + \frac{\Delta_t}{4} M^{-1}(p_+ + p_-).$$

Substitution into the momentum equation yields

$$\begin{aligned} E_{\Delta_t} &= \tfrac{\Delta_t^2}{4} K M^{-1} + \tfrac{\Delta_t}{2} C M^{-1}, \\ (I_n + E_{\Delta_t}) p_+ &= (I_n - E_{\Delta_t}) p_- - \Delta_t K q_- \\ &\quad + \Delta_t (F_m + U_m), \\ q_+ &= q_- + \tfrac{\Delta_t}{2} M^{-1}(p_+ + p_-). \end{aligned} \tag{S57}$$

The order KM^{-1} follows from substitution and cannot be interchanged with $M^{-1}K$. A state-dependent input still makes the update implicit. If $M = M(q)$, the momentum equation also contains the vector

$$(b_M)_i = -\tfrac{1}{2} \bar{p}^{\mathsf{T}} [\partial_{q_i} M^{-1}(\bar{q})] \bar{p}.$$

It is then preferable to solve the full $2n$-dimensional midpoint residual. This inertial contribution is not a scalar denominator correction.

In balanced coordinates the effective kinetic mass is $M_{\text{eff}} = \mathsf{N}^{-1}$, so $M_{\text{eff}}^{-1} Y = \mathsf{N} Y$ and proportional damping corresponds to $C_{\text{eff}} = \gamma \mathsf{N}^{-1}$. The mass in mass-normalized physical modal coordinates is instead I_n, while the assembled finite-element mass acts on all unconstrained beam degrees of freedom. These three matrices describe different coordinate systems.

S2.4 Exact symplecticity of the central map

Since S''_Δ is symmetric, $C_\Delta^{\mathsf{T}} J + J C_\Delta = 0$. Expanding the two products gives

$$(I - C_\Delta)^{\mathsf{T}} J (I - C_\Delta) = (I + C_\Delta)^{\mathsf{T}} J (I + C_\Delta).$$

The factors $I \pm C_\Delta$ commute, so (S56) gives $D\Psi_\Delta^{\mathsf{T}} J D\Psi_\Delta = J$. To check the stronger exact property, set $\delta X = X_+ - X_-$ and $\delta Y = Y_+ - Y_-$. The central relation says $\delta X = S_{\bar{Y}}$ and $\delta Y = -S_{\bar{X}}$. Therefore

$$\begin{aligned} Y_+^{\mathsf{T}}\,\mathrm{d}X_+ - Y_-^{\mathsf{T}}\,\mathrm{d}X_- &= \bar{Y}^{\mathsf{T}}\,\mathrm{d}(\delta X) + \delta Y^{\mathsf{T}}\,\mathrm{d}\bar{X} \\ &= \mathrm{d}(\bar{Y}^{\mathsf{T}}\delta X - S_\Delta). \end{aligned} \tag{S58}$$

The primitive is single valued wherever the branch and S_Δ are single valued. In the polar chart used for normal form,

$$\sum_j Y_j\,\mathrm{d}X_j = \sum_j I_j\,\mathrm{d}\phi_j - \mathrm{d}\sum_j I_j \sin\phi_j \cos\phi_j.$$

The difference is the differential of a periodic function. Subsequent normal-form transformations are Hamiltonian flows of single-valued generators. Consequently, the conjugated central map is exact symplectic also on the action–angle annulus, which is the global one-form condition needed for a periodic mixed generating function.

S2.5 Derivation of the fourth-order central generating function

Consider an autonomous smooth vector field $f = J\nabla h$, and let x_0 be the exact state at the temporal midpoint of a step.

Taylor expansion gives

$$\begin{aligned} m &= x_0 + \tfrac{\Delta^2}{8} f' f(x_0) + O(\Delta^4), \\ x_+ - x_- &= \Delta f(x_0) \\ &\quad + \tfrac{\Delta^3}{24}\{f''[f,f] + (f')^2 f\}(x_0) + O(\Delta^5). \end{aligned} \tag{S59}$$

Here m is the average of endpoint states, not the temporal midpoint. Eliminating x_0 yields

$$\begin{aligned} x_+ - x_- &= \Delta f(m) \\ &\quad + \tfrac{\Delta^3}{24}\{f''[f,f] - 2(f')^2 f\}(m) + O(\Delta^5). \end{aligned} \tag{S60}$$

For $A = h''$ and $Q = f^{\mathsf{T}} A f$, differentiation gives

$$\begin{aligned} \nabla Q &= h^{(3)}[f,f] - 2AJAf, \\ J\nabla Q &= f''[f,f] - 2(f')^2 f. \end{aligned}$$

This proves (36). On a fixed finite interval with bounded derivatives and a stable implicit branch, the local fifth-order defect gives fourth-order global convergence. Since $S^{[4]}_{-\Delta} = -S^{[4]}_{\Delta}$, the method is also symmetric under reversal of endpoints and step sign. These are properties of this autonomous Hamiltonian method; they do not assert fourth order for a midpoint discretization of a general time-dependent feedback law.

For $h = \nu(X^2 + Y^2)/2$, the central coefficient is $a = \nu\Delta + (\nu\Delta)^3/12$. Its angle advance in the positive-frequency convention is

$$2\arctan(a/2) = \nu\Delta - \frac{(\nu\Delta)^5}{120} + O(\Delta^7). \tag{S61}$$

Midpoint instead has angle $2\arctan(\nu\Delta/2)$ and a cubic local phase defect. This test distinguishes the two schemes without conflating exact symplecticity with exact energy or phase conservation for nonlinear systems.

S2.6 Explicit Central4 derivatives for the magnetic potential

For $h_\kappa = Y^{\mathsf{T}}\mathsf{N}Y/2 + V_\kappa(X)$, set $V_\kappa = X^{\mathsf{T}}\mathsf{N}X/2 + \kappa g(d^{\mathsf{T}}X)$, $u = \nabla V_\kappa$, $A = \nabla^2 V_\kappa$, $v = \mathsf{N}Y$, $s = d^{\mathsf{T}}X$ and $b = d^{\mathsf{T}}v$. Then $f^{\mathsf{T}} h''_\kappa f = v^{\mathsf{T}} A v + u^{\mathsf{T}}\mathsf{N}u$. Differentiation gives the directly implementable central increment

$$\begin{aligned} J\nabla S^{[4]}_{\Delta} &= \Delta \begin{pmatrix} v \\ -u \end{pmatrix} + \frac{\Delta^3}{24}\begin{pmatrix} Q \\ P \end{pmatrix}, \\ Q &= 2\mathsf{N}Av, \\ P &= \kappa\sin(s) b^2 d - 2A\mathsf{N}u. \end{aligned} \tag{S62}$$

The correction Jacobian has blocks

$$\begin{aligned} Q_X &= -2\kappa\sin(s) b(\mathsf{N}d)d^{\mathsf{T}}, \\ Q_Y &= 2\mathsf{N}A\mathsf{N}, \\ P_X &= \kappa[\cos(s) b^2 \\ &\quad + 2\sin(s) d^{\mathsf{T}}\mathsf{N}u] dd^{\mathsf{T}} - 2A\mathsf{N}A, \\ P_Y &= 2\kappa\sin(s) b\, d(\mathsf{N}d)^{\mathsf{T}}. \end{aligned} \tag{S63}$$

These formulas follow from $A = \mathsf{N} + \kappa(\cos s - 1)dd^{\mathsf{T}}$ and its derivative; they retain every cross-mode term. They apply to the specified autonomous conservative Hamiltonian. For a broader learned residual the complete derivatives of that residual must replace the single-feature formulas. A damping force is not included by adding it after this central update.

S2.7 Implicit solvability and accumulation of algebraic residuals

For the ordinary controlled midpoint equation, fix x_k and define $\mathcal{F}_k(z) = z - x_k - \Delta f_c((z + x_k)/2, \tau_m)$. On a convex local neighborhood where f_c has Lipschitz constant L_c, let $a = \Delta L_c/2 < 1$. The fixed-point map is a contraction when the step is small enough to map that neighborhood into itself. If z_* is its exact root, subtraction of $\mathcal{F}_k(z_*) = 0$ from $\mathcal{F}_k(z) = r$ yields

$$(1 - a)\,\|z - z_*\| \le \|r\|\,.$$

The same subtraction for two different inputs gives the stability factor $q_\Delta = (1 + a)/(1 - a)$ for the exact one-step map. For the exact and computed trajectories in Proposition 3, iteration gives the explicit weighted sum

$$\begin{aligned} e_N &\le q_\Delta^N e_0 \\ &\quad + \frac{1}{1 - a}\sum_{j=0}^{N-1} q_\Delta^{N-1-j}(C\Delta^3 + \eta_j). \end{aligned} \tag{S64}$$

With $N\Delta \le T$ and a bounded away from one, $q_\Delta^N \le e^{C_L T}$, giving the uniform bound in (50). For constant $L_c = 0$, the same formula holds with $q_\Delta = 1$. The result permits fixed nonzero δR and $\delta\gamma$: these change f_c and its constants, not the order of the midpoint approximation to that same field.

For a central generator satisfying $S''_\Delta = O(\Delta)$ on the neighborhood, the implicit derivative is $I - JS''_\Delta/2$. A uniform inverse bound and a one-step stability factor $1 + O(\Delta)$ give the same residual accumulation argument. The autonomous Central4 construction has a local defect $O(\Delta^5)$ under bounded derivatives of sufficient order, so

$$\max_{k\Delta \le T} e_k \le C_T\big(e_0 + \Delta^4 + \eta_\Delta/\Delta\big).$$

In particular, $O(\Delta^5)$ per-step residuals retain fourth order. Neither this estimate nor a small reported residual makes the inexact computed step an exact symplectic map.

The weighted method has the corresponding second-order finite-time result through its canonical lift. On a fixed interval, $T_\tau(X,Y) = (X, e^{\delta\gamma\tau}Y)$ and its inverse have bounded norms. Complete time-midpoint discretization of the lifted Hamiltonian obeys the preceding local and global argument, and transforming back changes only constants depending on those bounds. Residuals in the physical weighted equations must first be converted to the lifted increment equation; the conversion is bounded on that interval. This does not give a time-uniform error bound as T tends to infinity.

The residual r_k in (48) is an algebraic increment defect. Observation and quadrature defects in the identification equation are different quantities and enter the coefficient bound in (60). A bounded but nonconservative actuation error affects the realized vector field, while a zero-order hold changes the feedback map. These quantities are not added to an analytic Hamiltonian norm unless their conservative representation has first been justified.

S2.8 Analytic extension of a complex energy readout

For a real energy $h(X,Y)$, use $\zeta = (X + \mathrm{i}Y)/\sqrt{2}$ and $\chi = (X - \mathrm{i}Y)/\sqrt{2}$ as independent complex variables after continuation. The physical slice is $\chi = \bar{\zeta}$. If a frozen network expression $F(\zeta,\chi)$ is holomorphic on a domain invariant under $(\zeta,\chi) \mapsto (\bar{\chi},\bar{\zeta})$, define

$$F^\sharp(\zeta,\chi) = \overline{F(\bar{\chi},\bar{\zeta})}, \qquad h^{\mathbb{C}} = \tfrac{1}{2}(F + F^\sharp). \tag{S65}$$

Conjugating the frozen coefficients makes $F^\sharp$ holomorphic. Restriction to the physical slice gives $h = \operatorname{Re} F$, while differentiation off that slice is holomorphic differentiation. This construction explicitly separates a real-valued readout from a nonholomorphic operation on extended variables. A real-valued holomorphic function of ζ alone would be constant on a connected open complex domain, so that is not the required energy representation.

For a general physical force f in the momentum equation, the normalized complex dynamics and power identity are

$$\zeta' = -\mathrm{i}h^{\mathbb{C}}_\chi + \frac{\mathrm{i}}{\sqrt{2}}f, \qquad h' = h^{\mathsf{T}}_Y f \tag{S66}$$

for an autonomous energy. If a residual symbol is defined by $\zeta' = -\mathrm{i}h^{\mathbb{C}}_\chi + \mathcal{P}/2$, its normalization is $\mathcal{P} = \sqrt{2}\,\mathrm{i}f$. A scalar real potential that generates this arbitrary force through a second energy branch need not exist: the corresponding derivatives must satisfy the mixed-derivative compatibility conditions. Directly decoding the force or a specified Rayleigh dissipation avoids imposing an unwarranted gradient representation.

On a complex polydisc, Cauchy's formula bounds each first derivative on a smaller polydisc by the energy sup norm divided by the lost width. Thus a force norm can be inferred from an analytic energy bound after explicit domain reduction. The reverse implication from a finite sample of real gradients to a complex sup norm does not follow. The finite-feature estimate (62) supplies the needed extension through known basis functions.

S2.9 From linear canonical normalization to quartic normal form

At a quiescent stable equilibrium with $M_* \succ 0$ and potential Hessian $K_* \succ 0$, diagonalize $E^{\mathsf{T}} M_*^{-1/2} K_* M_*^{-1/2} E = \Omega^2$ with E orthogonal. The transformation

$$q - q_* = AQ, \qquad p = A^{-\mathsf{T}}P, \qquad A = M_*^{-1/2} E \Omega^{-1/2}$$

is canonical and gives $H_2 = \frac{1}{2}\sum_j \omega_j (Q_j^2 + P_j^2)$. This is the general source of the balanced quadratic form; physical mass and stiffness modes alone are not nonlinear action variables.

For $H = H_2 + H_3 + H_4 + \cdots$, solve $\{H_2, W_3\} = -H_3$ when the required cubic divisors are nonzero. Since the pullback by the time-one flow of W_3 is $H \circ \Phi^1_{W_3} = H + \{H, W_3\} + \frac{1}{2}\{\{H, W_3\}, W_3\} + \cdots$, the quartic term after removing H_3 is

$$Z_4 = \left\langle H_4 + \tfrac{1}{2}\{H_3, W_3\} \right\rangle. \tag{S67}$$

The nonaveraged quartic part is removed by a second homological equation. This formula fixes both the sign and the contribution of cubic terms; simply averaging the original H_4 would in general be insufficient. The generic remainder starts at degree five. In the present even magnetic model, $H_3 = 0$ and the remainder starts at degree six.

To compute its quartic term, expand

$$\begin{aligned} h_{4,\kappa} = -\frac{\kappa}{24}\big(&d_1^4 X_1^4 + 4 d_1^3 d_2 X_1^3 X_2 + 6 d_1^2 d_2^2 X_1^2 X_2^2 \\ &+ 4 d_1 d_2^3 X_1 X_2^3 + d_2^4 X_2^4\big). \end{aligned} \tag{S68}$$

Independent angle averages give $\langle X_j^4 \rangle = 3I_j^2/2$, $\langle X_1^2 X_2^2 \rangle = I_1 I_2$, and zero for the odd cross terms. Hence

$$Z_{4,\kappa} = -\frac{\kappa}{16}(d_1^4 I_1^2 + d_2^4 I_2^2) - \frac{\kappa}{4} d_1^2 d_2^2 I_1 I_2,$$

whose action Hessian is exactly A_κ in (65). In the convention (ϕ, I), $\{h_2, W\} = -\nu \cdot \partial_\phi W$. If $h_{4,\kappa} - Z_{4,\kappa} = \sum_{k\neq 0} P_{4,k}(I) e^{\mathrm{i}k\cdot\phi}$, a solution is

$$W_{4,\kappa} = \sum_{k\neq 0} \frac{P_{4,k}(I)}{\mathrm{i}k\cdot\nu} e^{\mathrm{i}k\cdot\phi}. \tag{S69}$$

Only finitely many divisors, with $|k|_1 \le 4$, occur. Written back in Cartesian coordinates this is a quartic polynomial, so its local flow is regular at the equilibrium. The polar chart is used only on an action annulus bounded away from zero, where branches of $\sqrt{I_j}$ have an analytic continuation.

For $|X_j| \le r$, set $z = \|d\|_1 r$. The original cosine tail obeys

$$|h_{\ge 6,\kappa}| \le |\kappa| \frac{z^6 \cosh z}{720}. \tag{S70}$$

The full transformed remainder also contains $\{h_{4,\kappa}, W_{4,\kappa}\}$ and $\frac{1}{2}\{\{h_2, W_{4,\kappa}\}, W_{4,\kappa}\}$, both of degree six, and higher Lie terms. A rigorous finite-amplitude bound must control the local flow of $W_{4,\kappa}$, its image and inverse, these brackets and the tail on a domain with margin. The notation $R_6 = O(r^6)$ states an order of vanishing, not a numerical sup-norm certificate. For the new baseline, evaluation of the retained finite divisors gives $\min_{0<|k|_1\le 4} |k\cdot\nu| = 1$; this does not bound the infinite family of divisors at nonlinear frequencies.

S2.10 The integral-error inequality and implementation uncertainty

For exact window integrals and endpoint-matching curves, the physical defect is

$$\begin{aligned} e_\ell = &\int_{a_\ell}^{b_\ell} [\dot c_{\ell Y} + \nabla V_0(c_{\ell X})]\,\mathrm{d}\tau \\ &+ \int_{a_\ell}^{b_\ell} [\textstyle\sum_j \beta_{*,j} \nabla F_j(c_{\ell X}) + \gamma c_{\ell Y} - U_\ell^{\mathrm{data}}]\,\mathrm{d}\tau. \end{aligned} \tag{S71}$$

This is exactly $y_\ell - D_\ell\theta_*$. If the true potential has a class residual R_{out}, the identity remains the algebraic definition of e_ℓ, and the omitted force contributes to its bound. Reconstructed trajectories need not solve the model; their deviation from the true trajectories is part of the observation uncertainty. When numerical quadrature is used, define $e = y - D\theta_*$ for the exported matrices and include their integration errors and coefficient-dependent design uncertainty in ϵ_I.

For completeness, let $\widetilde D, \widetilde y$ denote ideal matrices and suppose $\left\|D - \widetilde D\right\|_2 / \sqrt{nL_I} \le \epsilon_D$, $\|y - \widetilde y\|_2 / \sqrt{nL_I} \le \epsilon_y$, and $\left\|\widetilde y - \widetilde D\theta_*\right\|_2 / \sqrt{nL_I} \le \epsilon_{\mathrm{obs}}$. An a priori bound $\|\theta_*\|_2 \le B_\theta$ gives

$$\epsilon_I \le \epsilon_{\mathrm{obs}} + \epsilon_y + \epsilon_D B_\theta.$$

The Gram lower bound can be certified directly for the exported matrix, or transferred with singular-value perturbation estimates. If $\sigma_{\min}(\widetilde D/\sqrt{nL_I}) \ge s_0$, then $\lambda_{\min}(D^{\mathsf T}D/(nL_I)) \ge (s_0 - \epsilon_D)^2$ when $s_0 > \epsilon_D$. This prevents a quadrature-dependent excitation claim from being treated as exact.

The estimate (60) follows from

$$\sqrt{\underline\lambda}\left\|\widehat\theta - \theta_*\right\|_2 \le \frac{\left\|D(\widehat\theta - \theta_*)\right\|_2}{\sqrt{nL_I}} \le r_I + \epsilon_I.$$

It does not require least-squares optimality. If the reported residual is recomputed from the frozen exported coefficients, their storage rounding is already represented; time-stepping roundoff remains a separate dynamical implementation error. If observations are first corrected using $\widehat\gamma$, the corrected defect retains $-\delta\gamma \int c_{\ell Y}\,\mathrm{d}\tau$. Joint identification avoids losing this term by treating the estimated damping as known truth.

S2.11 A finite CHNN readout beyond the single-feature benchmark

Let $m_4(x)$ contain the $\binom{7}{4} = 35$ homogeneous quartic monomials in $x = (X_1, X_2, Y_1, Y_2)$, in a declared fixed order. Let v_4 be the coefficient vector of $-(d^{\mathsf T}X)^4/24$ in this basis and let $U_F \in \mathbb{R}^{35\times 34}$ have orthonormal columns spanning $v_4^\perp$. Define

$$\begin{aligned} F(x) &= \begin{pmatrix} g(d^{\mathsf T}X)/\left\|v_4\right\|_2 \\ U_F^{\mathsf T} m_4(x) \end{pmatrix}, \\ h_\eta(x) &= h_2(x) + \eta^{\mathsf T} F(x), \\ \eta &= (a\left\|v_4\right\|_2, c), \qquad c \in \mathbb{R}^{34}. \end{aligned} \tag{S72}$$

The quartic Taylor coefficients span all 35 directions, while the first feature retains the full cosine tail. The nominal energy has $a = 1, c = 0$; the minimal learned model has $a = 1 + \widehat\beta, c = 0$. Thus the present experiment is a restricted member of this analytic readout. The 35-coefficient model was not retrained on the new baseline, and no performance or certification result from an earlier parameter set is assigned to it.

If the broader state equation is modeled as $x' = L_2 x + V_F(x)\eta - \gamma Q(x) + u^{\mathrm{data}}$, where $L_2 x = J\nabla h_2$, $V_F = [J\nabla F_1, \dots, J\nabla F_{35}]$, $Q = (0, Y)^{\mathsf T}$ and $u^{\mathrm{data}} =$

$(0, U^{\mathrm{data}})^{\mathsf{T}}$, its integrated design is

$$D_\ell^{35} = \int_{a_\ell}^{b_\ell} [V_F(c_\ell), -Q(c_\ell)]\,\mathrm{d}\tau,$$
$$y_\ell^{35} = c_\ell(b_\ell) - c_\ell(a_\ell) - \int_{a_\ell}^{b_\ell} (L_2 c_\ell + u^{\mathrm{data}})\,\mathrm{d}\tau. \tag{S73}$$

Each window has four scalar equations and 36 unknown coefficients including damping. The normalizing count is $4L_I$, and the defect and Gram analysis must be repeated in this dimension. Window-length normalization is also possible, but must be applied to both D_ℓ and y_ℓ and propagated consistently to the norms. A certified coefficient bound then gives $\left\|\widehat{h} - h_*\right\|_{\mathcal{U}} \le b_{35}(\sum_{j=1}^{35} \|F_j\|_{\mathcal{U}}^2)^{1/2}$ plus a declared class-error budget.

The column Q assumes the stated linear momentum damping. If the general energy changes h_Y and the velocity-based Rayleigh law of (18) is retained, its force is $-\gamma \mathsf{N}^{-1} h_Y$ rather than $-\gamma Y$. The damping regressor must then be changed, and joint identification need not remain linear in the energy and damping coefficients.

Momentum-dependent learned terms change h_Y. They do not fit the direct pure-potential feedback laws (41) and (S84). To use such a model as a reference, a target momentum $\widetilde{Y}$ can instead be recovered from $\widehat{h}_Y(X, \widetilde{Y}) = \omega_0^{-1} D_s^{-1} \dot{a}$, provided $\widehat{h}_{YY}$ is locally invertible on the selected domain. The reference acceleration is

$$\ddot{a}^{\mathrm{ref}} = \omega_0^2 D_s (\widehat{h}_{YX}\widehat{h}_Y - \widehat{h}_{YY}\widehat{h}_X). \tag{S74}$$

The physical fixed-kinetic plant still requires an independently appropriate physical force estimate for compensation. A momentum term in a learned reference is not an estimate of an unknown physical potential. Complex and real implementations of the same finite readout should therefore be compared with matched features, data and error budgets.

S2.12 The map-KAM hypothesis in the central discretization

For the principal compensation problem take $\mathcal{T} = \mathcal{T}_1$, $H = h_c \circ \mathcal{T}_1 = H_{\mathrm{NF}}^1 + P_c$ and $\|P_c\| \le \epsilon_c$. The following argument also applies to a separately verified reference chart in Supplement S4; write $H = H_{\mathrm{NF}} + P$ and $\|P\| \le \epsilon$ for either declared case. The exact numerical map to analyze is $\widehat{\Psi}_\Delta = \mathcal{T}^{-1} \Psi_\Delta \mathcal{T}$. It is exact symplectic by Proposition 2 and Supplement S2.4. Analytic local error estimates on a smaller domain give $\left\|\widehat{\Psi}_\Delta - \Phi_H^\Delta\right\| \le C\Delta^{p+1}$. Cauchy bounds on ∇P and a short-time flow estimate give $\left\|\Phi_H^\Delta - \Phi_{H_{\mathrm{NF}}}^\Delta\right\| \le C'\Delta\epsilon$. Adding them proves the form of (68). The chart derivative and inverse-derivative bounds are part of these constants.

For a sufficiently small map perturbation the mixed endpoint projection $(I_-, \phi_-) \mapsto (I_+, \phi_-)$ is invertible. Exactness then supplies a periodic mixed generating perturbation G_Δ, fixed up to a constant, such that

$$I_+ = I_- - \partial_\phi G_\Delta(I_+, \phi_-),$$
$$\phi_+ = \phi_- + \Delta\omega(I_+) + \partial_I G_\Delta(I_+, \phi_-). \tag{S75}$$

Its domain-reduced bound is $\|G_\Delta\| \le C_3\Delta\epsilon + C_4\Delta^{p+1}$, the joint perturbation form used in the nearly integrable application of Shang (1999, Section 4). Constants are uniform only on a declared family of domains with the required analytic and inverse bounds.

For clarity, one sufficient theorem route is recorded explicitly. On an action complex neighborhood of radius ρ_I, suppose

$$\iota \|v\|_\infty \le \|D_I\omega(I)v\|_\infty \le \Theta \|v\|_\infty, \qquad 0 < \iota \le \Theta. \tag{S76}$$

For the affine frequency map these bounds follow from A and A^{-1}, for example $\iota = \left\|A^{-1}\right\|_\infty^{-1}$ and $\Theta = \|A\|_\infty$. Choose $\tau_D > n+1, 0 < \Delta \le 1$ and $0 < \gamma_D \le \min(1, \rho_I\Theta/2)$. The retained frequencies have distance at least $2\gamma_D$ from the frequency-domain boundary and satisfy (70). The analytic version of Shang (1999, Theorem 2 and Remark 2) requires, on its stated smaller complex action–angle domain,

$$\|G_\Delta/\Delta\| \le \delta_0 \gamma_D^2 \iota^2 \Theta^{-3}, \tag{S77}$$

where $\delta_0 > 0$ depends on the theorem's dimension, exponent and complex widths. A bounded frequency domain with the specified boundary regularity permits its positive-measure conclusion. For two modes we use the sufficient exponent $\tau_D > 3$ from this theorem. The lower-order continuous condition $\tau_D > 1$ in (S90) belongs to a different theorem.

Thus a verified inequality $C_3\epsilon + C_4\Delta^p \le \delta_0\gamma_D^2\iota^2\Theta^{-3}$, together with the domain and nonresonance conditions, provides the proposed discrete conclusion. The formula records what must be checked; it is not a numerical evaluation of δ_0, of the chart constants or of the complete remainder for the new baseline. It also shows why a small-step limit at fixed nonzero modeling perturbation does not remove that perturbation. The flow and discrete-map tori need not be identical.

S2.13 Finite-amplitude certification and scope of the hypotheses

An alternative to the local normal-form budget is an a posteriori certificate for a proposed torus. Given a periodic analytic embedding $K : \mathbb{T}^2 \to \mathbb{R}^4$ and a fixed frequency ω, its flow defect is

$$E(\phi) = J\nabla h(K(\phi)) - DK(\phi)\omega.$$

An applicable a posteriori theorem additionally requires a complex-strip extension, domain margins, nondegeneracy/torsion inverse bounds and a Diophantine constant; the defect must meet that theorem's quantitative inequality (Haro and Luque 2019). A numerical Fourier representation and a small sampled defect alone do not verify these premises. No previously computed embedding, remainder, correction radius or amplitude certificate is assigned to the new material and magnet model.

The principal chain connects a compatible generating construction, the actual compensated implicit equations, finite-time convergence with algebraic residuals, and conditional persistence of their exact conservative map. The hypotheses change at each step: finite-time convergence permits residual damping, whereas the classical map-KAM theorem does not. Reference tracking in Supplement S4 and full-beam actuation and clearance in Section 5 require their own control and mechanical estimates. Neither follows from a numerical invariant-torus diagnostic alone.

S3 Numerical protocol and result provenance

S3.1 Data records, numerical provenance and reproduction

This supplement specifies the experiments reported in Section 3. The supplied reproduction directory contains the finite-element generator, the two control scripts, a damping scan, the new framework verification scripts and their JSON records. The scalar inputs define a new numerical benchmark. Literature reference properties, manufacturer ranges, research design values and calculated outputs retain those distinct meanings throughout the study.

The recorded environment is Python 3.12.14, NumPy 2.3.5 and SciPy 1.17.0. From the root of the supplied source package, the complete numerical dependency chain can be regenerated by

```
cd reproduction
python run_all.py
```

The outer driver first rebuilds the FE baseline and then invokes the experiment driver. The latter executes the following stages in order:

1. `minimal_control_loop.py`;
2. `route_b_shared_data.py`;
3. `damping_sensitivity.py`;
4. `summarize_results.py`.

It sets the OpenBLAS, OpenMP and MKL thread counts to one. Route B requires the newly generated route-A result for a consistency check, while independently regenerating the shared observations and refitting their coefficients.

The numerical manuscript values and scientific figures are rendered directly from these existing JSON files by

```
cd reproduction/framework_experiments
python build_manuscript_assets.py
```

from the package root. This authoring script does not rerun the experiments. Its numerical-provenance JSON records the input hashes, source fields and displayed rounding. Floating-point results may change in their last digits across numerical libraries; bitwise identity is not used as a substitute for the reported convergence checks. The distributed audit also retains the hashes read by the actual runs and documents a subsequent annotation-only correction to the energy explanation and legacy damping metadata. The physical parameter, modal and sensitivity sections were unchanged by that correction.

S3.2 Complete physical inputs and coordinate conventions

The physical model is a uniform, solid, planar Euler–Bernoulli beam with a clamped root and an ideal rigid magnetic core at its free end. Its free length is $L = 0.08$ m, radius $r = 0.0006$ m, approximate Young modulus $E = 2.1 \times 10^6$ Pa and approximate density $\rho = 1028$ kg/m^3. The latter two numbers are a paired material reference near 10 wt% curing-agent fraction, with curing at 80 °C for 2 h (Cho et al. 2021). A 10 wt% fraction of the complete mixture corresponds approximately to a 9:1 base-to-curing-agent mass ratio. These data do not make the target round beam a measured specimen of that study.

The bare core is a design cube with edge $a = 0.00025$ m, nominal density $\rho_m = 7500$ kg/m^3 and remanence $B_r = 1.425$ T. The manufacturer range used for comparison is 1.40–1.45 T, and the density is selected at its stated lower bound (SM Magnetics 2026). The dimensional moment estimate is $\mu \simeq B_r a^3/\mu_0$, with $\mu_0 \simeq 4\pi \times 10^{-7}$ H/m (K&J Magnetics 2026). It is an engineering approximation to a uniformly

Table S3. Derived nominal properties. Values come from `new_baseline.json`; displayed precision describes the calculation, not experimental precision.

Quantity	Calculated value	Unit
Area A	$1.13097336 \times 10^{-6}$	m^2
Second moment I	$1.01787602 \times 10^{-13}$	m^4
Flexural rigidity EI	$2.13753964 \times 10^{-7}$	$\mathrm{N\,m}^2$
Beam mass ρAL	$9.30112487 \times 10^{-5}$	kg
Core volume V_m	$1.56250000 \times 10^{-11}$	m^3
Core mass m_t	$1.17187500 \times 10^{-7}$	kg
Core inertia J_t	$1.22070312 \times 10^{-15}$	$\mathrm{kg\,m}^2$
Dipole moment μ	$1.77184214 \times 10^{-5}$	$\mathrm{A\,m}^2$
Energy scale E_0	$5.31552642 \times 10^{-7}$	J

magnetized hard magnet. The core centre is assumed to coincide with the beam endpoint; its magnetization is tangent to the straight reference beam. The constant longitudinal field is $B_0 = 0.03$ T. The model has no transverse gravity, contact, external flow, distributed beam rotary inertia, shear deformation, coating, adhesive or fixture mass. Each omission is a modeling assumption.

The dimensional modal basis satisfies $\Phi^{\mathsf{T}} M_{\mathrm{total}} \Phi = I$. Its balanced displacement scale is $S_j = \sqrt{E_0/(\omega_0 \omega_j)}$, where $E_0 = \mu B_0$ and $\omega_0 = \omega_1$. Thus the physical modal displacement and momentum are $a_j = S_j X_j$ and $\pi_j = \sqrt{E_0 \omega_j / \omega_0}\, Y_j$. In the dimensionless time $\tau = \omega_0 t$, $X' = \mathsf{N} Y$ and the effective mass in the second-order X equation is N^{-1}. It is not the identity after balanced scaling. Endpoint angle and displacement are $d^{\mathsf{T}} X$ and $\ell^{\mathsf{T}} X$, with the coefficients of (53). This same map is used for all physical error metrics.

S3.3 Finite-element and independent boundary checks

Each cubic-Hermite element uses the usual consistent translational mass and bending stiffness. Root displacement and slope are removed, leaving $2n_e$ configuration coordinates. The core adds m_t to the terminal displacement mass and $J_t = m_t a^2/6$ to the terminal rotation inertia. The bias adds μB_0 to the terminal rotational stiffness. The eigensolve is mass normalized and uses a scaled inverse symmetric eigenproblem to resolve the lowest modes without subtracting them from the high-frequency spectral end. The first four frequencies are used for the grid check; the first two define the control model.

The independent continuum check solves $EIw'''' = \rho A \omega^2 w$ with $w(0) = w'(0) = 0$ and

$$\begin{aligned} EIw'''(L) + m_t \omega^2 w(L) &= 0, \\ EIw''(L) + (\mu B_0 - J_t \omega^2) w'(L) &= 0. \end{aligned} \tag{S78}$$

The clamped basis uses $\cosh(kx) - \cos(kx)$ and $\sinh(kx) - \sin(kx)$. Its endpoint determinant is bracketed separately and solved by Brent iteration with absolute root tolerance 2×10^{-13}. This is an independent floating-point check of the same ideal boundary model, not an interval eigenvalue proof.

Table S5 changes one declared input at a time. The core-mass factor multiplies both m_t and J_t, holding geometry and moment fixed. It is an independent inertia sensitivity, not a manufacturer tolerance or a fully coupled manufacturing model. In particular, the 0.9 factor is not claimed to represent a same-size core within the adopted density specification. The 20% modulus changes are also designed perturbations. Four combined remanence/inertia corners are retained in the JSON. Throughout these FE parameter sensitivities the nominal dimensional α is held fixed; the resulting damping ratio therefore changes with frequency. This convention differs deliberately from the separate damping scan, which fixes the physical baseline and changes α.

S3.4 Training data, validation data and the integral fit

The nominal synthetic plant uses $\beta_* = 0.2$ and $\gamma_* = 0.05$. Its complete cosine is retained. Training and validation states are generated with classical RK4 step 0.0005 over $0 \le \tau \le 4$. The physical duration of each record is 0.510203 s; it is not four seconds. Sampling every fourth generator step gives 2001 states per trajectory, at observation spacing 0.002. No random noise, smoothing or derivative estimation is applied. The six fixed initial states are given in Table S6. They are a new, lower-amplitude design for this softer beam; each is one half of the corresponding initial state in the preceding minimal experiment. They are not samples from a measured distribution.

Each nonoverlapping window contains 20 intervals and 21 states. Adjacent windows share their boundary observation. Composite Simpson weights are applied directly to the two integrals in (56). Each trajectory supplies 100 vector windows: the four training trajectories give 800 scalar rows, and the two validation trajectories give 400 scalar rows. The two design columns are divided by their Euclidean norms, solved by an unconstrained least-squares routine, and rescaled. There is no regularization or nonnegative constraint. The source script verifies full column rank.

The normalized joint Gram matrix $G = A^{\mathsf{T}} A/800$ is

$$G = \begin{pmatrix} 5.16502121 \times 10^{-14} & 2.38379831 \times 10^{-11} \\ 2.38379831 \times 10^{-11} & 5.59508836 \times 10^{-6} \end{pmatrix}. \tag{S79}$$

Table S4. Finite-element frequencies in rad/s and maximum relative error across the first four modes against the independent continuum roots. The 32-element line defines the adopted baseline.

Elements	ω_1	ω_2	ω_3	ω_4	Maximum relative error
16	7.84001022	47.39980944	131.21158140	256.37539514	1.5071×10^{-4}
32	7.84000916	47.39957662	131.20668107	256.33922193	9.5903×10^{-6}
64	7.84000910	47.39956199	131.20637133	256.33691793	6.0208×10^{-7}

Table S5. Selected FE design sensitivities. The final column uses the fixed nominal $\alpha = 0.3920004581\ \mathrm{s}^{-1}$. None is a measured confidence interval.

Perturbation	ω_1 (rad/s)	Change (%)	$\alpha/(2\omega_1)$
$B_r = 1.40$ T	7.83350702	−0.08294	0.02502075
$B_r = 1.45$ T	7.84648941	+0.08266	0.02497935
m_t, J_t scaled by 0.9	7.84192087	+0.02438	0.02499391
m_t, J_t scaled by 1.1	7.83809879	−0.02437	0.02500609
E scaled by 0.8	7.09311163	−9.52674	0.02763247
E scaled by 1.2	8.51964699	+8.66884	0.02300567

Table S6. Dimensionless initial states. The validation trajectories are excluded from parameter fitting.

Record	X_1	X_2	Y_1	Y_2
Training 1	0.09	0.07	0.025	−0.02
Training 2	−0.075	0.1	0.01	0.025
Training 3	0.11	−0.06	−0.015	0.03
Training 4	0.05	0.12	0.02	−0.01
Validation 1	−0.095	0.055	0.03	0.005
Validation 2	0.065	−0.105	−0.02	0.015

Here the normalization uses scalar rows; replacing it by the number of vector windows would change the numerical constant in a coefficient bound. With $r_I = \|A\widehat{\theta} - b\|_2/\sqrt{800}$, an independently justified defect bound $\|A\theta_* - b\|_2/\sqrt{800} \le \varepsilon_I$ would give the conditional estimate described in Section 4. Using the known synthetic truth as a diagnostic yields $\varepsilon_{I,\mathrm{diag}} = 1.26876 \times 10^{-12}$ and a bound expression 1.00470×10^{-5}, compared with actual coefficient error 3.36849×10^{-6}. The diagnostic uses privileged truth and ordinary floating-point quadrature. It is not an independent enclosure of ε_I and cannot close a rigorous analytic-norm or KAM argument. The held-out residual evaluates different trajectories within the same feature family; it does not test a larger feature class or a noisy observation model.

S3.5 Route-A protocol and midpoint verification

The initial state is $(0.09, 0.08, 0.015, -0.01)$ and the target is the conservative flow of h_0. Four command choices are evaluated: $U = 0$, $U = \widehat{\gamma}Y$, $U = \widehat{\gamma}Y + \widehat{\beta}\phi(X)$, and $U = \gamma_* Y + \beta_*\phi(X)$. The last is the oracle. Commands are reconstructed through the same ideal physical-port routine and the reconstructed values enter the plant differential equation. Thus port inversion is exercised during integration rather than only evaluated after a modal simulation.

Table S7. Route-A oracle midpoint convergence to $\tau = 20$. The maximum is over all saved common times.

Δ	Maximum state error	Observed order
0.04	4.64310×10^{-2}	—
0.02	1.18409×10^{-2}	1.971306
0.01	2.96760×10^{-3}	1.996414
0.005	7.42245×10^{-4}	1.999331

The principal midpoint step is 0.001, the final time is $\tau = 20$, and every endpoint is stored, giving 20001 states. The physical time step is 1.27551×10^{-4} s, with no claim that it is a realizable hardware sampling period. An explicit-Euler predictor initializes each Newton solve. The stopping criterion is

$$\left\| x_{k+1} - x_k - \Delta f\left(\frac{x_k + x_{k+1}}{2}\right)\right\|_\infty \le 2 \times 10^{-14}. \quad \text{(S80)}$$

The analytic midpoint Jacobian is used, and the loop raises an error after 12 unsuccessful correction attempts. This criterion concerns the full increment equation rather than the correction norm alone. The recorded principal route-A solves used at most one correction; the coarser order-check runs used at most two.

The independent h_0 reference is computed by RK4 at steps 0.0005 and 0.00025. The fine reference is sampled at every fourth step for the principal comparison. The oracle midpoint convergence test uses steps 0.04, 0.02, 0.01 and 0.005, always comparing at common times with that independent fine reference.

Observed orders are $\log_2(E_\Delta/E_{\Delta/2})$. The symplectic diagnostic uses the spectral norm of $D\Psi^{\mathsf{T}} J D\Psi - J$, where $D\Psi$ is the ideal implicit Cayley tangent evaluated at the solved midpoint. Only the oracle restoration is assigned this diagnostic as a test of a Hamiltonian target map. The

supplied matrix-elimination check separately uses $n = 1, 2, 4$ configuration coordinates to verify the algebraic second-order update; it is not a high-mode beam-accuracy experiment.

S3.6 Damping scan with newly generated observations

The five first-mode ratios are 0, 0.01, 0.025, 0.05 and 0.1. For each, the dimensional drag is $\alpha = 2\zeta_1^{\mathrm{d}}\omega_0$ and the true dimensionless drag is $\gamma = 2\zeta_1^{\mathrm{d}}$. The FE basis, artificial $\beta_* = 0.2$ and all initial states remain fixed. All training and validation observations are regenerated, and both coefficients are refitted by the unchanged unconstrained two-column procedure. Thus no value is obtained by copying the baseline fit or changing only the controller parameter.

The restoration route is tested with midpoint step 0.002 until $\tau = 10$, or 1.275509 s. Its independent h_0 reference uses RK4 step 0.00025. Passive, learned and oracle groups are included; the damping-only group of the principal test is not repeated. All errors here use the full $[0, 10]$ window. The supplementary tracking extension uses the same refitted parameters but its own target and late window; those distinct results are retained in Table S16. The principal-run convergence checks do not constitute a separate convergence study at every scan point.

S3.7 Metric definitions, physical ports and sampled extrema

For saved errors $e_k = x_k - x_{\mathrm{ref}}(\tau_k)$, $k = 0, \dots, n$, the state metric is

$$E_{\mathrm{RMS}} = \left(\frac{1}{n+1} \sum_{k=0}^{n} \|e_k\|_2^2 \right)^{1/2}. \tag{S81}$$

It is a Euclidean error in four balanced dimensionless coordinates; there is no further division by four. Late-window RMS applies the same definition to the indicated subset of saved states. The maximum state error is $\max_k \|e_k\|_2$ and the endpoint error is $\|e_n\|_2$. Physical tip displacement errors use $\ell^{\mathsf{T}} e_{X,k}$, and endpoint-angle errors use $d^{\mathsf{T}} e_{X,k}$, before the scalar RMS or maximum is taken. The additional tracking experiment retains its separate independent and synchronous reference metrics in Supplementary Section S4.5.

Route-A energy drift is $\max_k |h_0(x_k) - h_0(x_0)|$, with a relative version divided by $|h_0(x_0)|$. This is nominal energy, not an exact nonlinear action or the energy of every perturbed plant. Command effort is the trapezoidal approximation to $\int \|U\|_2^2 \, \mathrm{d}\tau$; it is dimensionless generalized effort, not electrical actuator power. The regression residual is a different metric: scalar-row RMSE of $A\widehat{\theta} - b$.

The local port reconstruction assumes independently commandable force and torque with exact state information. The field and gradient reconstruction is substituted back into the modal equation, and its generalized-force difference is recorded. The total local field is $\sqrt{B_0^2 + B_y^2}$ and the local gradient size is the spectral norm of its symmetric, trace-free two-dimensional matrix. No actuator-current map or saturation is implemented.

Trajectory maxima for angle and displacement are sampled at saved times. Whole-beam slopes and displacements use Hermite interpolation on a 257-point spatial grid, eight subdivisions per element. The nodal shapes are supplied, so spatial polynomial extrema can be checked separately, but the reported figures use the stated grid and do not enclose time-continuous extrema. Command diagnostics include all endpoints and arithmetic midpoints used by the midpoint integrator. The 20 mT transverse-field value discussed as a design comparison is not an enforced clipping rule.

S3.8 P1 framework matrix verification

The additional experiments are stored separately in `reproduction/framework_experiments`. From the package root, their complete entry point is

```
cd reproduction/framework_experiments
python run_all.py
```

The optional argument `--experiment` selects `matrix`, `closed`, `defect` or `weighted`. All parameter paths are resolved relative to the scripts. Existing FE, training and restoration records are read and not overwritten. The new JSON files contain the baseline and source-code hashes and the numerical-library versions. The accompanying authoring script renders the current three numerical inputs, including the supplementary tracking extension; the older two-route authoring templates are archival files.

For the dimensionless matrix family in (52), the initial states, ordered as (q, p), are

$$\begin{aligned} n = 1: &\quad (0.1, 0), \\ n = 2: &\quad (0.1, 0.06, 0, 0.02), \\ n = 4: &\quad (0.1, 0.06, 0.04, 0.025, 0, 0.02, -0.01, 0.005). \end{aligned}$$

Each dimension uses $T = 4$ and the four steps $0.08, 0.04, 0.02, 0.01$. SciPy's matrix exponential evaluates $\exp(tA)x_0$ independently at all 401 points of the finest grid. Every coarser method is compared at its own subset of those times. The ordinary map and its eliminated form are advanced

Table S8. New fit in every damping scenario. The tiny negative zero-damping estimate is numerical and is not constrained away.

ζ_1^{d}	$\alpha\ (\mathrm{s}^{-1})$	$\widehat{\beta}$	$\widehat{\gamma}$	$\lambda_{\min}(G)$
0	0.000000000	0.200003040949	$-1.4037896716 \times 10^{-12}$	6.9451×10^{-14}
0.01	0.156800183	0.200003177492	$1.9999999994 \times 10^{-2}$	6.1346×10^{-14}
0.025	0.392000458	0.200003368491	$4.9999999987 \times 10^{-2}$	5.1549×10^{-14}
0.05	0.784000916	0.200003636621	$9.9999999975 \times 10^{-2}$	3.9870×10^{-14}
0.1	1.568001832	0.200003938495	$1.9999999995 \times 10^{-1}$	2.6883×10^{-14}

Table S9. Direction-one damping scenarios, each with newly generated observations and a new fit. All entries are state RMS against the independent h_0 reference over $[0, 10]$.

ζ_1^{d}	Passive	Learned restoration	Oracle restoration
0	3.77362×10^{-5}	3.42885×10^{-5}	3.42885×10^{-5}
0.01	6.73657×10^{-3}	3.42885×10^{-5}	3.42885×10^{-5}
0.025	1.59501×10^{-2}	3.42885×10^{-5}	3.42885×10^{-5}
0.05	2.92396×10^{-2}	3.42885×10^{-5}	3.42885×10^{-5}
0.1	4.97374×10^{-2}	3.42885×10^{-5}	3.42885×10^{-5}

independently; their maximum trajectory discrepancy is saved for every dimension and step. A minimum singular value of $I + E_\Delta$ is also recorded.

The weighted map is $S_{1/2}\,\mathrm{Mid}_\Delta(A_0)S_{1/2}$, where $S_{1/2} = \mathrm{diag}(I_n, e^{-0.2\Delta/2}I_n)$. The conservative checks use $K + K_s$ with $K_s = 0.05\,\mathrm{diag}(1, \ldots, n)$, corresponding to the declared input $U = 0.2p - K_s q$. For a linear conservative field Ax, Central4 uses increment matrix $G_\Delta = \Delta A - \Delta^3 A^3/12$ and update $(I - G_\Delta/2)^{-1}(I + G_\Delta/2)$, evaluated by linear solves. The extra C_g family checks elimination only; nonproportional damping occurs for $n = 2, 4$, while the scalar member is a comparison case.

S3.9 P3 fixed-error closed field and conservative Central4

The added magnetic tests read the same `new_baseline.json` and use initial state $(0.09, 0.08, 0.015, -0.01)$ over $\tau \in [0, 20]$. The fixed-error experiment sets $\delta\beta = 0.02$, $\delta\gamma = 0.01$ on the plant with $\beta_* = 0.2$ and $\gamma_* = 0.05$, hence the declared controller coefficients are 0.18 and 0.04. They are intentionally manufactured errors and are not presented as a new identification result. No coefficient is refitted when the step changes. The midpoint right-hand side calls the existing physical port reconstruction inside every solve. The independent reference implements the reduced field (58) directly, so the reference does not reuse the midpoint algorithm.

Independent DOP853 references use a coarse setting $\mathrm{rtol} = 10^{-11}$, $\mathrm{atol} = 10^{-13}$, maximum step 0.05, and a fine setting $\mathrm{rtol} = 2 \times 10^{-13}$, $\mathrm{atol} = 2 \times 10^{-15}$, maximum step 0.025. Dense outputs are evaluated at the same saved times. Both success flags are checked. These are floating-point refinement checks, not rigorous solution enclosures. The midpoint steps are $0.04, 0.02, 0.01, 0.005, 0.0025$. The complete actual, nominal-reference and closed-reference trajectories are stored in `closed_loop_convergence_traces.npz`.

The nonlinear Central4 test instead fixes the conservative h_0 ($\kappa = 1$, zero residual drag) and uses the first four of those steps. It implements the complete increment and analytic Jacobian in (S62)–(S63). Complex-step checks with perturbation $10^{-25}i$ are made at the main initial state and $(-0.07, 0.04, 0.03, -0.02)$, using $\Delta = 0.04$. The maximum derivative discrepancy is 2.7756×10^{-17}. This test does not assert that the imperfectly compensated physical loop is Hamiltonian, or that appending a feedback kick preserves fourth order.

All implicit framework solves use an increment predictor, residual infinity tolerance 2×10^{-14} and at most 12 Newton corrections; failure stops the run. For midpoint the predictor is explicit Euler; for Central4 it is the complete increment evaluated at the previous state. Every run records the maximum actual residual, maximum correction count, failed-step count, and minimum sampled singular value of the Newton matrix. The standard and exponential-weighted defects evaluate the ideal implicit tangent at computed midpoints, not the derivative of the finite floating-point iteration. The general norm statement in (50) applies with consistent norms; for the magnetic four-state model an infinity residual bounds the Euclidean residual by at most twice that value.

Table S10. Sampled magnetic demands in the principal learned-restoration test. Endpoints and integration midpoints are included. These are ideal local demands, not measured actuator capabilities.

Test	$\max\lvert B_y\rvert$ (mT)	$\max\lvert B\rvert$ (mT)	$\max\lVert\nabla B\rVert_2$ (T/m)	$\max\lvert F_y\rvert$ (μN)
A: learned	0.544490	30.004941	0.01603429	0.284102

Table S11. P1 matrix full-grid maximum errors at $\Delta = 0.01$. Each column uses its own declared damped or conservative target. All preceding step errors and observed orders are retained in the JSON.

n	Ordinary midpoint	Weighted midpoint	Conservative midpoint	Conservative Central4
1	1.98924×10^{-6}	1.97226×10^{-6}	3.41418×10^{-6}	3.35319×10^{-11}
2	3.71730×10^{-6}	3.85870×10^{-6}	5.58365×10^{-6}	8.05545×10^{-11}
4	6.08098×10^{-6}	5.58130×10^{-6}	9.62132×10^{-6}	1.95812×10^{-10}

S3.10 Manufactured algebraic-defect budget

This independent one-degree-of-freedom test uses

$$A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad x_0 = (0.1, 0.02)^{\mathsf{T}}, \quad T = 4, \tag{S82}$$

and steps $0.08, 0.04, 0.02, 0.01$. It deliberately solves

$$\begin{aligned} x_{k+1} - x_k - \Delta A \frac{x_k + x_{k+1}}{2} &= \eta_\Delta (1, 0)^{\mathsf{T}}, \\ \eta_\Delta &= 0.01 \Delta^q, \end{aligned} \tag{S83}$$

for $q = 1, 2, 3$. The update is the ordinary midpoint matrix plus $(I - \Delta A/2)^{-1} \eta_\Delta (1, 0)^{\mathsf{T}}$. Thus the residual is introduced intentionally into the algebraic equation; it is not attributed to a Newton solver. The JSON contains every actual residual vector and infinity norm, its maximum Euclidean norm, and the maximum discrepancy from the prescribed residual. The latter is 3.5779×10^{-17} across the entire test.

Two errors are kept separate: the total maximum error against $\exp(tA)x_0$, and the added maximum error against an independently advanced unperturbed midpoint trajectory at the same step. The added error isolates the η_Δ/Δ term. The total error also contains the ordinary $O(\Delta^2)$ integration term.

S3.11 Nonlinear exponential-weighted comparison

The weighted experiment uses two explicitly known fields: the uncompensated synthetic plant $(\delta\beta, \delta\gamma) = (0.2, 0.05)$, and the fixed-error diagnostic $(0.02, 0.01)$. It uses the same initial state as the principal magnetic experiment, $T = 4$, and steps $0.04, 0.02, 0.01, 0.005$. For each field, ordinary midpoint and the weighted map are compared with independent DOP853 references at the settings above. The weighted implementation is $S_{1/2}\,\mathrm{Mid}_{h_\kappa}\,S_{1/2}$, with $S_{1/2} = \mathrm{diag}(I_2, e^{-\delta\gamma\Delta/2} I_2)$ and $\kappa = 1 + \delta\beta$. Its numerical solutions are also substituted back into (45). The weighted ideal tangent is $S_{1/2} D\,\mathrm{Mid}_{h_\kappa}\,S_{1/2}$, so the target factor is $e^{-\delta\gamma\Delta}$. The two models and all errors are recorded separately; no unknown true error is supplied to the deployed learned controller.

S3.12 Conditional clearance and historical results that do not transfer

The 4 mm clearance is a net centreline allowance after subtraction of the rod radius. The nominal energy comparison uses $E_c = 3EIc^2/(2L^3) = 10.019717$ nJ and the selected factor 0.95. It defines a conditional sufficient bound through the nonnegative nominal kinetic, elastic and static-field energies. The principal initial energy is 12.656721 nJ, exceeding the proposed 9.518731 nJ cap. The present controls do not implement a governor. A sampled displacement below the clearance therefore cannot be reported as verification of that energy-cap premise.

The earlier manuscript's angle, rate, velocity and actuator-error budgets were already declared deterministic engineering hypotheses. They remain such hypotheses, rather than being reclassified as measured sensor specifications. Their operational reserves and any delay bounds must be recalculated for the new physical maps before being used in a guarantee.

In particular, the old 22.66 nJ energy budget is not a new-baseline input. The old torus frequency, Fourier coefficients, 35-parameter fits, COMSOL trajectories, 200 s prediction statistics and full-order tracking errors describe different numerical objects. Their provenance can remain in the archival register, but none is presented as an output of the calculations reported here. The recomputed two-mode tests and the retained conditional theorems therefore have explicitly different evidential roles.

S4 Related extension: tracking a learned conservative reference

The principal manuscript studies residual compensation and the resulting discrete map. This extension retains the

Table S12. Manufactured defect diagnostic at $\Delta = 0.01$. Final observed orders compare steps 0.02 and 0.01. "Added" means relative to unperturbed midpoint, while "total" means relative to the continuous oscillator.

q	Actual residual, max. ∞ norm	Total error	Total order	Added order	Added error$/(\eta_\Delta/\Delta)$
1	1.000×10^{-4}	2.00026×10^{-2}	0.000566	-0.000000	1.999999
2	1.000×10^{-6}	2.02630×10^{-4}	1.018776	1.000000	1.999999
3	1.000×10^{-8}	5.18612×10^{-6}	1.999969	2.000000	1.999999

Table S13. Nonlinear weighted-map diagnostic. Defects are maximum exponential-weighted spectral-norm defects at $\Delta = 0.04$; orders use the last step halving. Both methods approximate the same field within each row pair.

Field $(\delta\beta, \delta\gamma)$	Ordinary defect	Weighted defect	Ordinary order	Weighted order
$(0.2, 0.05)$	2.8763×10^{-5}	3.3771×10^{-16}	1.999452	1.999400
$(0.02, 0.01)$	5.7618×10^{-6}	3.3603×10^{-16}	1.999414	1.999403

alternative task of tracking a learned conservative reference. Both tasks use the same frozen identification and physical input ports, but their target trajectories and mathematical guarantees differ.

S4.1 A learned conservative reference and tracking feedback

Route B instead generates a reference by $X_r' = \mathsf{N} Y_r$, $Y_r' = -\nabla \widehat{V}(X_r)$, where $\widehat{V} = V_0 + \widehat{R}$. With $e_X = X - X_r$ and $e_Y = Y - Y_r$, use

$$\begin{aligned} U_B = {} & \widehat{\gamma} Y + \nabla \widehat{V}(X) - \nabla \widehat{V}(X_r) \\ & - \kappa_p \mathsf{N}^{-1} e_X - \kappa_d e_Y, \qquad \kappa_p, \kappa_d > 0. \end{aligned} \tag{S84}$$

The error satisfies the exact equations

$$\begin{aligned} e_X' &= \mathsf{N} e_Y, \\ e_Y' &= -\kappa_p \mathsf{N}^{-1} e_X - \kappa_d e_Y \\ &\quad - \nabla \delta R(X) - \delta\gamma Y, \\ e_X'' &+ \kappa_d e_X' + \kappa_p e_X \\ &= -\mathsf{N}[\nabla \delta R(X) + \delta\gamma Y]. \end{aligned} \tag{S85}$$

The homogeneous dynamics are exponentially stable. On a declared domain where the force error and velocity are bounded, convolution with this stable system gives a finite tracking bound; the critically damped formula is in Supplement S4.2. Nonzero residual forcing gives a nonzero error envelope. Exact cancellation gives asymptotic tracking and preserves a zero initial tracking error.

For the synthetic truth-only comparator, define $V_* = V_0 + R_*$ and replace only the plant compensation terms:

$$\begin{aligned} U_B^{\mathrm{or}} = {} & \gamma Y + \nabla V_*(X) - \nabla \widehat{V}(X_r) \\ & - \kappa_p \mathsf{N}^{-1} e_X - \kappa_d e_Y. \end{aligned} \tag{S86}$$

This oracle retains the same learned reference (X_r, Y_r) as (S84); it does not replace that reference by the true conservative flow. Substitution gives the homogeneous error equation $e_X'' + \kappa_d e_X' + \kappa_p e_X = 0$. Its midpoint update is the Cayley transform of the constant error-system matrix, providing the numerical oracle check in Supplement S4.4. The implemented learned feedback remains (S84) and does not read R_* or γ.

KAM analysis in this route concerns $\widehat{h}$ and hence the existence of selected quasiperiodic references. The feedback system itself includes damping and a time-dependent reference. Its tracking guarantee comes from (S85), not from symplecticity of that closed loop. The magnetic port map in Section 5 applies to this two-mode generalized input under the same invertibility and actuation conditions. The reference, tracking and plant-compensation errors remain separate quantities.

S4.2 Tracking with a bounded model and actuation error

Let an additive realized-input error be r_U in the momentum equation. Then the Route B position error satisfies

$$\begin{aligned} e_X'' + \kappa_d e_X' + \kappa_p e_X &= q(\tau), \\ q &= -\mathsf{N}[\nabla \delta R(X) + \delta\gamma Y - r_U]. \end{aligned}$$

For $\kappa_p = \omega_c^2$, $\kappa_d = 2\omega_c$, direct solution gives

$$\begin{aligned} e_X(\tau) = {} & e^{-\omega_c \tau}[(1 + \omega_c \tau) e_X(0) + \tau e_X'(0)] \\ & + \int_0^\tau (\tau - s) e^{-\omega_c(\tau - s)} q(s)\, \mathrm{d}s. \end{aligned} \tag{S87}$$

If the plant and reference stay in the asserted domain and $\|q(s)\|_2 \le Q$, then

$$\begin{aligned} \|e_X(\tau)\|_2 \le {} & e^{-\omega_c \tau}[(1 + \omega_c \tau)\, \|e_X(0)\|_2 + \tau\, \|e_X'(0)\|_2] \\ & + \frac{Q}{\omega_c^2}[1 - (1 + \omega_c \tau) e^{-\omega_c \tau}]. \end{aligned} \tag{S88}$$

Table S14. Disposition of principal numerical claims after changing the material inputs, endpoint inertia and modal maps. The complete chapter-level register accompanies the source package.

Earlier component	Current treatment	Work required for a new numerical claim
Beam energy, complex representation and control identities	Retain derivations with the new mass and coordinate maps	Check all specialized assumptions and derivatives for the selected residual and ports.
FE frequencies and balanced maps	Recomputed at $16/32/64$ elements	Higher-mode forced trajectories remain a separate validation task.
Matched real/complex CHNN fits and training-seed contrasts	Historical; replaced here by a declared two-scalar test	Generate new data and perform the complete matched training and validation protocol.
GC17 midpoint/Central4 table	Historical numerical values excluded	Refit the intended energy and repeat both integration tests on that function.
COMSOL full-FE, two-mode and model-mismatch comparisons	Historical numerical values excluded	Rebuild each independent model, its inputs, solver protocol and reduction comparisons.
Finite-amplitude KAM, family/measure and tail certificates	Conditional theory retained; old numerical certificates excluded	Recompute the fitted function, analytic domains, Gram/defect bounds, twist, torus and validated remainder inequalities.
Full-beam torus lift and persistent-control bounds	Conditional construction; no new full-order simulation	Recompute higher modes, tail forcing, input bounds and startup transients.
Governor, operational tolerance and S-path results	Conditional arguments; old outputs excluded	Reinitialize and implement the corresponding controllers and reserves for the new model.
Long-time and resonance-boundary diagnostics	Historical outputs excluded	Rerun the stated horizons, sampling, continuation and convergence studies.

The velocity error follows by differentiating the convolution and using $e_Y = \mathsf{N}^{-1} e'_X$. In particular, if $\|Y\| \le Y_{\max}$, $\|r_U\| \le r_{\max}$, and (62) holds, one can take

$$Q = \|\mathsf{N}\|_2 \left[b_I B_{\nabla F}(K) + b_I Y_{\max} + \sup_K \|\nabla R_{\mathrm{out}}\|_2 + r_{\max} \right].$$

The two-mode tip displacement error is at most $\|\ell(L)\|_2 \, \|e_X\|_2$. These inequalities apply for as long as their domain and actuation assumptions hold. A bootstrap using a sufficiently large reference tube can establish self-consistency when its margin exceeds the right-hand side; otherwise domain exit must be checked separately. They are not a global result in the presence of arbitrary saturation or unmodeled modes.

S4.3 Reference persistence and its relation to the compensation budget

For Route A, choose the nominal chart $\mathcal{T}_A = \mathcal{T}_1$. For Route B in the single-feature model, choose $\mathcal{T}_B = \mathcal{T}_{\kappa_*}$ with $\kappa_* = 1 + \beta_*$. A broader true residual requires a corresponding normal form, or inclusion of its additional terms in the perturbation budget. For the single-feature plant, the energies relevant to the two claims satisfy

$$\begin{aligned} h_c \circ \mathcal{T}_A &= H^1_{\mathrm{NF}} + R^1_6 + \delta R \circ \mathcal{T}_A, \\ \widehat{h} \circ \mathcal{T}_B &= H^{\kappa_*}_{\mathrm{NF}} + R^{\kappa_*}_6 - \delta R \circ \mathcal{T}_B, \\ \epsilon_A &= \left\| R^1_6 \right\|_{\mathcal{D}_A} + b^A_R, \\ \epsilon_B &= \|R^{\kappa_*}_6\|_{\mathcal{D}_B} + b^B_R. \end{aligned} \tag{S89}$$

Here b^s_R bounds $\delta R \circ \mathcal{T}_s$ on the declared complex action–angle domain $\mathcal{D}_s$. A common coefficient bound can be reused, but a common analytic function bound is valid only if both chart images lie inside the same controlled physical complex neighborhood. The two normal forms generally have different frequencies, remainders and admissible action ranges.

For the synthetic plant $\kappa_* = 1.2$. For an unknown plant, (60) gives a parameter interval about $\widehat{\beta}$; the twist and remainder estimates must hold uniformly over the asserted interval, including exclusion of $\kappa_* = 0$. Alternatively, the frozen learned Hamiltonian can be put directly into normal form and certified on its own. Neither procedure identifies a physical invariant torus from a training loss alone.

Proposition S2. Conditional persistence for the continuous models. *For either route, let $\mathcal{T}_s$ be analytic and canonical on a complex neighborhood $\mathcal{D}_s$ of $\mathcal{I}_s \times \mathbb{T}^2$, with $\mathcal{I}_s \Subset (0,\infty)^2$. Assume that its real image lies in the modeled working domain, that the integrable frequency map $\omega_s = \nabla_I H^s_{\mathrm{NF}}$ is nondegenerate there, and that all the domain and norm*

bounds in (S89) are valid. Select an interior frequency set satisfying

$$|k \cdot \omega| \geq \gamma_D |k|_1^{-\tau_D}, \quad k \in \mathbb{Z}^2 \setminus \{0\}, \quad \tau_D > 1. \tag{S90}$$

If ϵ_s is below the analytic KAM threshold for these domains, twist bounds and Diophantine parameters, the corresponding Hamiltonian has a Cantor family of invariant Lagrangian tori carrying quasiperiodic flow. A positive-measure family follows for the interior domains and sufficiently small exclusion parameter in the classical theorem. For Route A this is a statement about the actual closed loop only under the conservative realization premise following (42); for Route B it is a statement about the learned reference.

The proposition follows by substituting (S89) into the analytic near-integrable Hamiltonian theorem (Pöschel 2001). Its threshold depends on the chosen complex widths, frequency-domain margins, twist and inverse-twist bounds, and γ_D, τ_D. No fixed numerical threshold is asserted here. The resulting tori describe selected bounded motions; they do not imply attraction of arbitrary initial conditions or clearance of the entire flexible beam.

S4.4 Route B: prediction, tracking and reference synchronization

Before control is applied, conservative prediction is tested against h_* using separate fine-step RK4 trajectories from the same initial state. The nominal h_0 state RMS is 8.04072×10^{-6}; replacing it by the fitted $\widehat{h}$ gives 1.35426×10^{-10}. This comparison tests recovery of the chosen conservative perturbation. The control experiment then freezes $\widehat{h}$, initializes its reference at the route-A initial state, and offsets the actual plant by $(0.0045, 0.004, 0, 0)$. Gains $\kappa_p = 1$ and $\kappa_d = 2$ make the position-error dynamics of the oracle law in (S86) critically damped. The same frozen reference is supplied to all four control groups, including the oracle.

The actual and reference states are integrated as one eight-dimensional midpoint system. A second, independently integrated RK4 reference is retained solely for assessment. Let x_k be the actual state, $\bar{x}_k$ the synchronous midpoint reference and $x_{\mathrm{ref}}(\tau_k)$ the independent reference. Then

$$x_k - x_{\mathrm{ref}}(\tau_k) = (x_k - \bar{x}_k) + (\bar{x}_k - x_{\mathrm{ref}}(\tau_k)). \tag{S91}$$

The first term measures synchronization to the implemented reference; the second retains its integration error. The two are reported together in Table S15.

The learned controller has whole-window independent-reference RMS 1.58912×10^{-3}, including the prescribed startup error. On $10 \leq \tau \leq 20$ it decreases to 2.26767×10^{-5}, while the synchronous RMS is 8.55100×10^{-7}. At the endpoint, the synchronous error is 2.98323×10^{-10} but the independent-reference error is 2.96937×10^{-5}. The maximum midpoint-reference error is 2.96936×10^{-5}; halving the midpoint step reduces that error by a factor 3.999984. Thus near-perfect synchronization is compatible with a resolved second-order reference phase error. It must not be reported as the accuracy of the continuous reference motion.

An additional check compares the oracle tracking error against the Cayley update of its linear error equation. The maximum discrepancy is 1.0040×10^{-15}. This verifies the augmented midpoint coupling without making the stable tracking loop Hamiltonian. The experiment also shows why damping compensation alone is insufficient for this objective: it does not remove the imposed initial position offset.

S4.5 Route-B prediction and tracking protocol

The route-B script regenerates the same training and validation data and checks its newly fitted coefficient vector against the route-A vector within 10^{-13}. It then evaluates three conservative trajectories, for h_0, $\widehat{h}$ and h_*, using RK4 step 0.00025 from the common initial state to $\tau = 20$. The prediction errors compare h_0 and $\widehat{h}$ with h_*; the latter is synthetic ground truth, not independent physical data. The $\widehat{h}$ and h_* references are each checked against an RK4 calculation at step 0.0005.

For feedback, the reference is the frozen conservative $\widehat{h}$ flow, initialized at $(0.09, 0.08, 0.015, -0.01)$. The actual initial state is $(0.0945, 0.084, 0.015, -0.01)$. The position offset is 5% of each reference position coordinate, with zero momentum offset. The gains are $\kappa_p = 1$, $\kappa_d = 2$ in dimensionless time. Passive, fitted damping-only, fitted tracking and oracle tracking groups use this same reference. The learned and oracle commands are the distinct laws in (S84) and (S86); the oracle uses true plant coefficients for cancellation but retains the fitted reference trajectory.

The four actual-state variables and four reference variables form an eight-dimensional midpoint solve with the same Newton settings as (S80). The independent RK4 trajectory is not fed into that controller. It is used afterward to report the integration component in (S91). A separate run halves the midpoint step to 0.0005; the actual-state maximum common-time difference is 2.22701×10^{-5} and the reference error falls by the factor 3.999984. A finite-difference check of the analytic augmented Jacobian has

Table S15. Route-B state RMS at $\Delta = 0.001$. "Independent" refers to fine-step RK4 for the same frozen $\widehat{h}$; "synchronous" refers to the reference component of the augmented midpoint system.

Control	Independent, $[0, 20]$	Independent, $[10, 20]$	Synchronous, $[10, 20]$
Passive	2.55875×10^{-2}	3.41583×10^{-2}	3.41580×10^{-2}
Damping compensation	6.02185×10^{-3}	6.02249×10^{-3}	6.02060×10^{-3}
Learned tracking	1.58912×10^{-3}	2.26767×10^{-5}	8.55100×10^{-7}
Oracle tracking	1.58912×10^{-3}	2.26767×10^{-5}	8.55090×10^{-7}

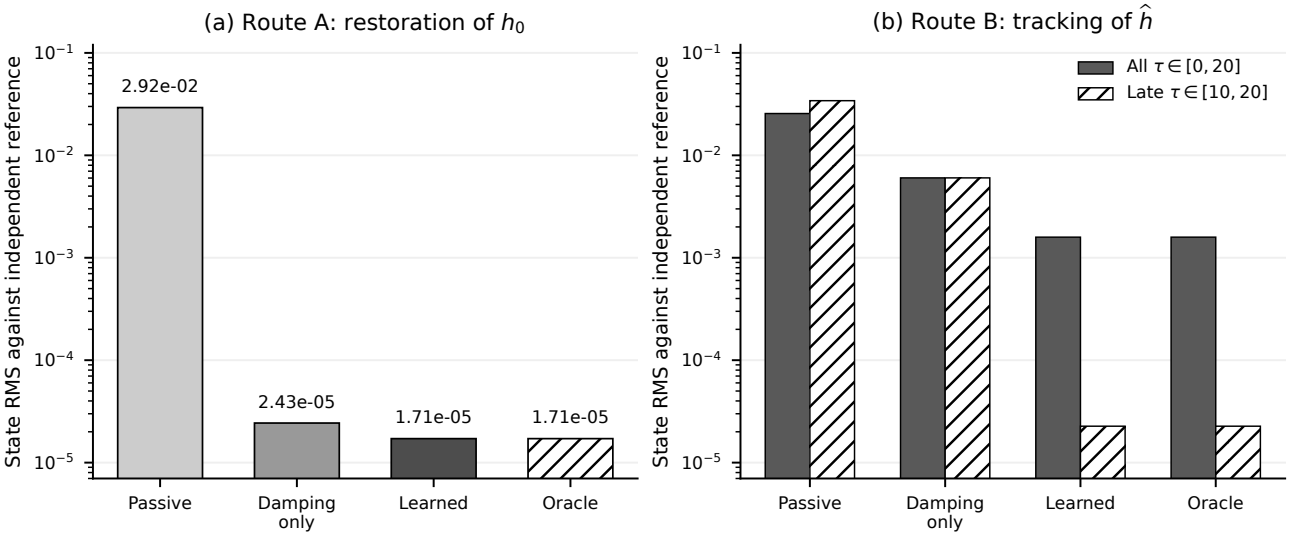


Figure S1. New-baseline comparisons using independent conservative references. (a) Route A restores h_0 from the same initial state. (b) Route B tracks a frozen $\widehat{h}$ trajectory from a nonzero initial offset. Whole-window and late-window errors expose the tracking transient. The panels compare different reference objects; their bars are not a ranking of one route against the other.

maximum discrepancy 3.6902×10^{-11} in the recorded check. These finite differences are an implementation diagnostic, not a replacement for the derived Jacobian.

For the oracle, the initial momentum-error vector is zero and the continuous error satisfies $e_X(\tau) = (1 + \tau)e^{-\tau}e_X(0)$ and $e_Y(\tau) = -\tau e^{-\tau}\mathsf{N}^{-1}e_X(0)$. The discrete error is also compared with the independently formed Cayley update of that linear system. The comparison tests the feedback and reference coupling with a known error equation.

S4.6 Tracking sensitivity and combined archived displays

The five damping settings use the same fresh observations and fits as the restoration tests in Supplementary Section S3.6. Their tracking runs last $\tau = 10$, with midpoint step 0.002 and independent RK4 step 0.00025; the late tracking window is $[5, 10]$. These are independent fits at each declared damping value, not one frozen controller exposed to unknown changing damping. The following table retains both targets to make their distinct time windows explicit.

Table S16. Independent-reference state RMS in the damping scan. The two routes use distinct target Hamiltonians and time windows.

ζ_1^{d}	A passive, $[0, 10]$	A learned, $[0, 10]$	B passive, $[5, 10]$	B learned, $[5, 10]$
0	3.77362×10^{-5}	3.42885×10^{-5}	6.02444×10^{-3}	1.09567×10^{-4}
0.01	6.73657×10^{-3}	3.42885×10^{-5}	3.71321×10^{-3}	1.09567×10^{-4}
0.025	1.59501×10^{-2}	3.42885×10^{-5}	1.62063×10^{-2}	1.09567×10^{-4}
0.05	2.92396×10^{-2}	3.42885×10^{-5}	3.42497×10^{-2}	1.09567×10^{-4}
0.1	4.97374×10^{-2}	3.42885×10^{-5}	6.13727×10^{-2}	1.09567×10^{-4}

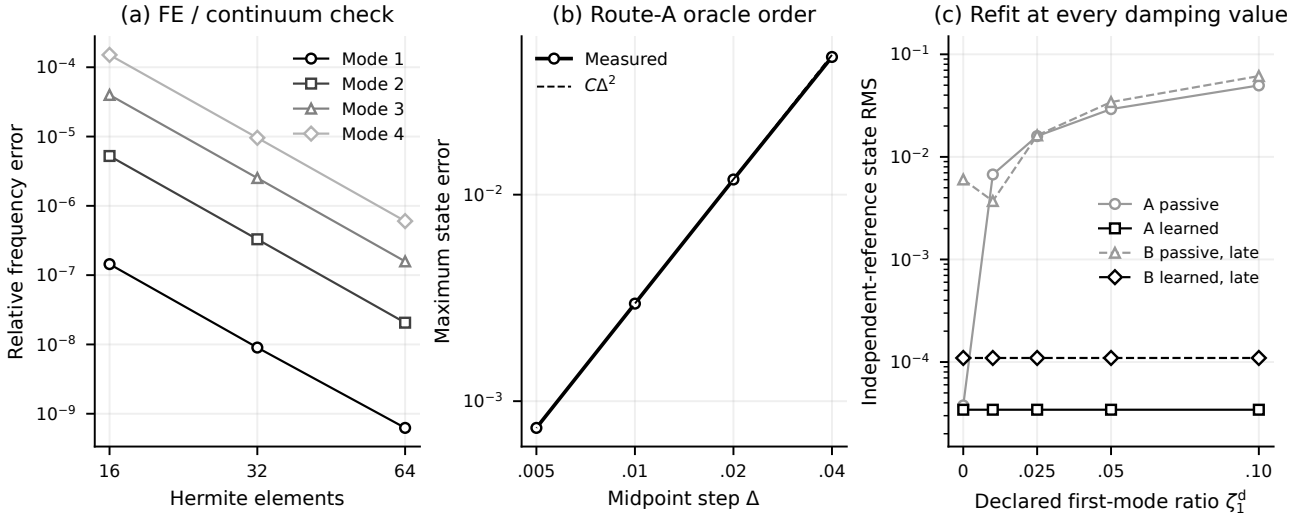


Figure S2. Independent numerical checks from the supplied JSON records. (a) The first four FE frequencies converge to roots of the continuum endpoint determinant. (b) Route-A oracle errors converge at second order against an independent RK4 reference. (c) Each damping setting is refitted before either route is evaluated; A uses the full $[0, 10]$ window and B the late $[5, 10]$ window. Lines connect tested settings and do not represent additional simulations or statistical intervals.

Table S17. Sampled local magnetic demand for the supplementary learned tracking test. This is an ideal command requirement, not a hardware capability measurement.

Test	$\max \lvert B_y \rvert$ (mT)	$\max \lvert B \rvert$ (mT)	$\max \lVert \nabla B \rVert_2$ (T/m)	$\max \lvert F_y \rvert$ (μN)
B: learned	2.197941	30.080408	0.03453753	0.611950

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